



? The higher the temperature of a gas, the greater the average kinetic energy of its molecules. How much faster are molecules moving in the air above a frying pan (100°C) than in the surrounding kitchen air (25°C)? (i) 4 times faster; (ii) twice as fast; (iii) 1.25 times as fast; (iv) 1.12 times as fast; (v) 1.06 times as fast.

18 Thermal Properties of Matter

The kitchen is a great place to learn about how the properties of matter depend on temperature. When you boil water in a tea kettle, the increase in temperature produces steam that whistles out of the spout at high pressure. If you forget to poke holes in a potato before baking it, the high-pressure steam produced inside the potato can cause it to explode messily. Water vapor in the air can condense into liquid on the sides of a glass of ice water; if the glass is just out of the freezer, water vapor will solidify and form frost on its sides.

These examples show the relationships among the large-scale or *macroscopic* properties of a substance, such as pressure, volume, temperature, and mass. But we can also describe a substance by using a *microscopic* perspective. This means investigating small-scale quantities such as the masses, speeds, kinetic energies, and momenta of the individual molecules that make up a substance.

The macroscopic and microscopic descriptions are intimately related. For example, the (microscopic) forces that occur when air molecules strike a solid surface (such as your skin) cause (macroscopic) atmospheric pressure. To produce standard atmospheric pressure of $1.01 \times 10^5 \text{ Pa}$, 10^{32} molecules strike your skin every day with an average speed of over 1700 km/h (1000 mi/h)!

In this chapter we'll begin by looking at some macroscopic aspects of matter in general. We'll pay special attention to the *ideal gas*, one of the simplest types of matter to understand. We'll relate the macroscopic properties of an ideal gas to the microscopic behavior of its molecules. We'll also use microscopic ideas to understand the heat capacities of gases and solids. Finally, we'll look at the various phases of matter—gas, liquid, and solid—and the conditions under which each occurs.

18.1 EQUATIONS OF STATE

Quantities such as pressure, volume, temperature, and amount of substance describe the conditions, or *state*, in which a particular material exists. (For example, a tank of medical oxygen has a pressure gauge and a label stating the volume within the tank. We can add a thermometer and put the tank on a scale to measure the mass of oxygen.) These quantities are called **state variables**.

LEARNING OUTCOMES

In this chapter, you'll learn...

- 18.1** How to relate the pressure, volume, and temperature of a gas.
- 18.2** How the interactions between the molecules of a substance determine the properties of the substance.
- 18.3** How the pressure and temperature of a gas are related to the kinetic energy of its molecules.
- 18.4** How the heat capacities of a gas reveal whether its molecules are rotating or vibrating.
- 18.5** How the speeds of molecules are distributed in a gas.
- 18.6** What determines whether a substance is a gas, a liquid, or a solid.

You'll need to review...

- 7.4** Potential energy and force.
- 11.4** Bulk stress.
- 12.2** Fluids in equilibrium.
- 13.3** Escape speed.
- 14.4** Interatomic forces and oscillations.
- 17.1–17.6** Temperature, heat, thermal expansion, specific heat, molar heat capacity, phase changes.

The volume V of a substance is usually determined by its pressure p , temperature T , and amount of substance, described by the mass m_{total} or number of moles n . (We are calling the total mass of a substance m_{total} because later in the chapter we'll use m for the mass of one molecule.) Ordinarily, we can't change one of these variables without causing a change in another. When the tank of oxygen gets hotter, the pressure increases. If the tank gets too hot, it explodes.

In a few cases the relationship among p , V , T , and m_{total} (or n) is simple enough that we can express it as an equation called the **equation of state**. When it's too complicated for that, we can use graphs or numerical tables. Even then, the relationship among the variables still exists; we call it an equation of state even when we don't know the actual equation.

Here's a simple (though approximate) equation of state for a solid material. The temperature coefficient of volume expansion β (see Section 17.4) is the fractional volume change $\Delta V/V_0$ per unit temperature change, and the compressibility k (see Section 11.4) is the negative of the fractional volume change $\Delta V/V_0$ per unit pressure change. If a certain amount of material has volume V_0 when the pressure is p_0 and the temperature is T_0 , the volume V at slightly differing pressure p and temperature T is approximately

$$V = V_0[1 + \beta(T - T_0) - k(p - p_0)] \quad (18.1)$$

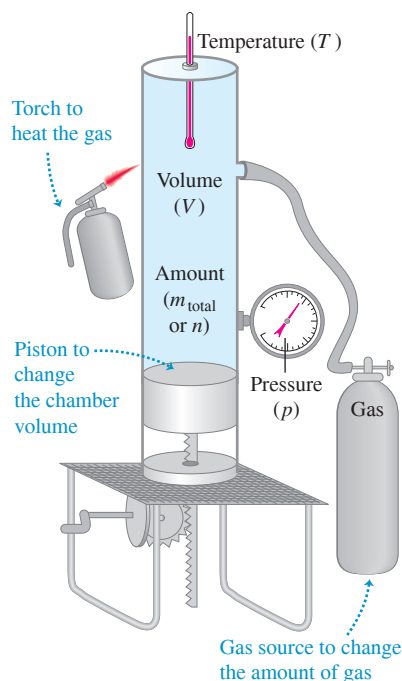
(There is a negative sign in front of the term $k(p - p_0)$ because an *increase* in pressure causes a *decrease* in volume.)

The Ideal-Gas Equation

Another simple equation of state is the one for an *ideal gas*. **Figure 18.1** shows an experimental setup to study the behavior of a gas. The cylinder has a movable piston to vary the volume, the temperature can be varied by heating, and we can pump in any desired amount of gas. We then measure the pressure, volume, temperature, and amount of gas. Note that *pressure* refers both to the force per unit area exerted by the cylinder on the gas and to that exerted by the gas on the cylinder; by Newton's third law, these must be equal.

It is usually easiest to describe the amount of gas in terms of the number of moles n , rather than the mass. (We did this when we defined molar heat capacity in Section 17.5.) The **molar mass** M of a compound (sometimes confusingly called *molecular weight*) is the mass per mole:

Figure 18.1 A hypothetical setup for studying the behavior of gases. By heating the gas, varying the volume with a movable piston, and adding more gas, we can control the gas pressure p , volume V , temperature T , and number of moles n .



$$\text{Total mass of substance} \rightarrow m_{\text{total}} = nM \left(\begin{array}{l} \text{Number of moles} \\ \text{of substance} \\ \text{Molar mass of substance} \end{array} \right) \quad (18.2)$$

Hence if we know the number of moles of gas in the cylinder, we can determine the mass of gas from Eq. (18.2).

Measurements of the behavior of various gases lead to three conclusions:

1. The volume V is proportional to the number of moles n . If we double n , keeping pressure and temperature constant, the volume doubles.
2. The volume varies *inversely* with the absolute pressure p . If we double p while holding the temperature T and number of moles n constant, the gas compresses to one-half of its initial volume. In other words, $pV = \text{constant}$ when n and T are constant.
3. The pressure is proportional to the *absolute* temperature T . If we double T , keeping the volume and number of moles constant, the pressure doubles. In other words, $p = (\text{constant}) \times T$ when n and V are constant.

We can combine these three relationships into a single **ideal-gas equation**:

$$\text{Ideal-gas equation: } pV = nRT \quad (18.3)$$

Gas pressure $\rightarrow p$ Gas volume $\rightarrow V$ Number of moles of gas $\rightarrow n$ Absolute temperature of gas $\rightarrow T$ Gas constant $\rightarrow R$

An **ideal gas** is one for which Eq. (18.3) holds precisely for *all* pressures and temperatures. This is an idealized model; it works best at very low pressures and high temperatures, when the gas molecules are far apart and in rapid motion. It is valid within a few percent at moderate pressures (such as a few atmospheres) and at temperatures well above those at which the gas liquefies (**Fig. 18.2**).

CAUTION Use absolute pressure and absolute temperature in the ideal-gas equation The pressure p in Eq. (18.3), the ideal-gas equation, is the *absolute* pressure, not the gauge pressure (see Section 12.2). Furthermore, the temperature T in Eq. (18.3) is the *absolute* (Kelvin) temperature, not the Celsius temperature. If you use gauge pressure or Celsius temperature in Eq. (18.3), you'll get nonsensical results, such as the pressure of a gas being negative below $0^\circ\text{C} = 273.15\text{ K}$!

We might expect that the proportionality constant R in the ideal-gas equation would have different values for different gases, but it turns out to have the same value for *all* gases, at least at sufficiently high temperature and low pressure. It is called the **gas constant** (or *ideal-gas constant*). In SI units, in which the unit of p is Pa ($1\text{ Pa} = 1\text{ N/m}^2$) and the unit of V is m^3 , the numerical value of R (to nine significant figures) is

$$R = 8.31446262\text{ J/mol} \cdot \text{K}$$

or $R = 8.314\text{ J/mol} \cdot \text{K}$ to four significant figures. Note that the units of pressure times volume are the same as the units of work or energy (for example, N/m^2 times m^3); that's why R has units of energy per mole per unit of absolute temperature. In chemical calculations, volumes are often expressed in liters (L) and pressures in atmospheres (atm). In this system, to four significant figures,

$$R = 0.08206 \frac{\text{L} \cdot \text{atm}}{\text{mol} \cdot \text{K}}$$

We can express the ideal-gas equation, Eq. (18.3), in terms of the mass m_{total} of gas, using $m_{\text{total}} = nM$ from Eq. (18.2):

$$pV = \frac{m_{\text{total}}}{M}RT \quad (18.4)$$

From this we can get an expression for the density $\rho = m_{\text{total}}/V$ of the gas:

$$\rho = \frac{pM}{RT} \quad (18.5)$$

CAUTION Density vs. pressure When using Eq. (18.5), be certain that you distinguish between the Greek letter ρ (rho) for density and the letter p for pressure.

For a *constant mass* (or constant number of moles) of an ideal gas the product nR is constant, so the quantity pV/T is also constant. If the subscripts 1 and 2 refer to any two states of the same mass of a gas, then

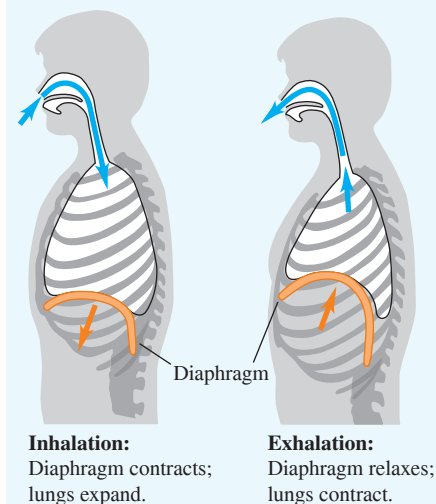
$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2} = \text{constant} \quad (\text{ideal gas, constant mass}) \quad (18.6)$$

Notice that you don't need the value of R to use this equation.

Figure 18.2 The ideal-gas equation $pV = nRT$ gives a good description of the air inside an inflated vehicle tire, where the pressure is about 3 atmospheres and the temperature is much too high for nitrogen or oxygen to liquefy. As the tire warms (T increases), the volume V changes only slightly but the pressure p increases.



BIO APPLICATION Respiration and the Ideal-Gas Equation To breathe, you rely on the ideal-gas equation $pV = nRT$. Contraction of the dome-shaped diaphragm muscle increases the volume V of the thoracic cavity (which encloses the lungs), decreasing its pressure p . The lowered pressure causes the lungs to expand and fill with air. (The temperature T is kept constant.) When you exhale, the diaphragm relaxes, allowing the lungs to contract and expel the air.



We used the proportionality of pressure to absolute temperature in Chapter 17 to define a temperature scale in terms of pressure in a constant-volume gas thermometer. That may make it seem that the pressure–temperature relationship in the ideal-gas equation, Eq. (18.3), is just a result of the way we define temperature. But the ideal-gas equation also tells us what happens when we change the volume or the amount of substance. Also, we'll see in Chapter 20 that the gas-thermometer scale corresponds closely to a temperature scale that does *not* depend on the properties of any particular material. For now, consider Eq. (18.6) as being based on this genuinely material-independent temperature scale.

PROBLEM-SOLVING STRATEGY 18.1 Ideal Gases

IDENTIFY *the relevant concepts:* Unless the problem states otherwise, you can use the ideal-gas equation to find quantities related to the state of a gas, such as pressure p , volume V , temperature T , and/or number of moles n .

SET UP *the problem* using the following steps:

1. List the known and unknown quantities. Identify the target variables.
2. If the problem concerns only one state of the system, use Eq. (18.3), $pV = nRT$ (or Eq. (18.5), $\rho = pM/RT$ if the problem involves the density ρ rather than n and V).
3. In problems that concern two states (call them 1 and 2) of the same amount of gas, if all but one of the six quantities p_1 , p_2 , V_1 , V_2 , T_1 , and T_2 are known, use Eq. (18.6), $p_1V_1/T_1 = p_2V_2/T_2 = \text{constant}$. Otherwise, use Eq. (18.3) or Eq. (18.5).

EXECUTE *the solution* as follows:

1. Use consistent units. (SI units are entirely consistent.) The problem statement may make one system of units more convenient

than others. Make appropriate unit conversions, such as from atmospheres to pascals or from liters to cubic meters.

2. You may have to convert between mass m_{total} and number of moles n , using $m_{\text{total}} = Mn$, where M is the molar mass. If you use Eq. (18.4), you *must* use the same mass units for m_{total} and M . So if M is in grams per mole (the usual units for molar mass), then m_{total} must also be in grams. To use m_{total} in kilograms, you must convert M to kg/mol. For example, the molar mass of oxygen is 32 g/mol or 32×10^{-3} kg/mol.
3. Remember that in the ideal-gas equations, T is always an *absolute* (Kelvin) temperature and p is always an absolute (not gauge) pressure.
4. Solve for the target variables.

EVALUATE *your answer:* Do your results make physical sense? Use benchmarks, such as the result of Example 18.1 below that a mole of an ideal gas at 1 atmosphere pressure occupies a volume of 22.4 liters.

EXAMPLE 18.1 Volume of an ideal gas at STP

WITH VARIATION PROBLEMS

What is the volume of a container that holds exactly 1 mole of an ideal gas at *standard temperature and pressure* (STP), defined as $T = 0^\circ\text{C} = 273.15\text{ K}$ and $p = 1\text{ atm} = 1.013 \times 10^5\text{ Pa}$?

IDENTIFY and SET UP This problem involves the properties of a single state of an ideal gas, so we use Eq. (18.3). We are given the pressure p , temperature T , and number of moles n ; our target variable is the corresponding volume V .

EXECUTE From Eq. (18.3), using R in J/mol · K, we get

$$V = \frac{nRT}{p} = \frac{(1\text{ mol})(8.314\text{ J/mol} \cdot \text{K})(273.15\text{ K})}{1.013 \times 10^5\text{ Pa}} = 0.0224\text{ m}^3 = 22.4\text{ L}$$

EVALUATE At STP, 1 mole of an ideal gas occupies 22.4 L. This is the volume of a cube 0.282 m (11.1 in.) on a side, or of a sphere 0.350 m (13.8 in.) in diameter.

KEYCONCEPT The ideal-gas equation, Eq. (18.3), relates the pressure, volume, absolute temperature, and number of moles for a quantity of an ideal gas.

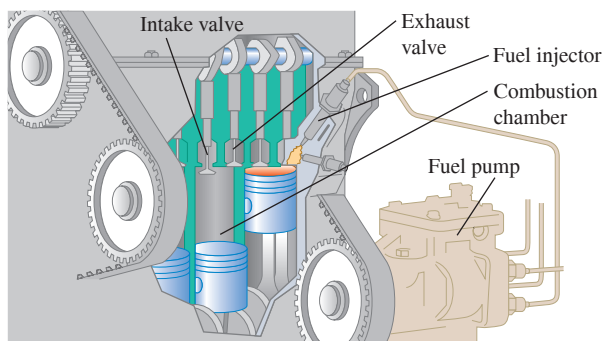
EXAMPLE 18.2 Compressing gas in an automobile engine

WITH VARIATION PROBLEMS

In an automobile engine, a mixture of air and vaporized gasoline is compressed in the cylinders before being ignited. A typical engine has a compression ratio of 9.00 to 1; that is, the gas in the cylinders is compressed to $\frac{1}{9.00}$ of its original volume (**Fig. 18.3**). The intake and exhaust valves are closed during the compression, so the quantity of gas is constant. What is the final temperature of the compressed gas if its initial temperature is 27°C and the initial and final pressures are 1.00 atm and 21.7 atm, respectively?

IDENTIFY and SET UP We must compare two states of the same quantity of ideal gas, so we use Eq. (18.6). In the uncompressed state 1, $p_1 = 1.00\text{ atm}$ and $T_1 = 27^\circ\text{C} = 300\text{ K}$. In the compressed state 2, $p_2 = 21.7\text{ atm}$. The cylinder volumes are not given, but we have $V_1 = 9.00V_2$. The temperature T_2 of the compressed gas is the target variable.

Figure 18.3 Cutaway of an automobile engine. While the air–gasoline mixture is being compressed prior to ignition, both the intake and exhaust valves are in the closed (up) position.



EXECUTE We solve Eq. (18.6) for T_2 :

$$T_2 = T_1 \frac{p_2 V_2}{p_1 V_1} = (300 \text{ K}) \frac{(21.7 \text{ atm}) V_2}{(1.00 \text{ atm})(9.00 V_2)} = 723 \text{ K} = 450^\circ\text{C}$$

EVALUATE This is the temperature of the air–gasoline mixture *before* the mixture is ignited; when burning starts, the temperature becomes higher still.

KEYCONCEPT For a fixed quantity of an ideal gas, the pressure p , volume V , and absolute temperature T may all change, but the quantity pV/T remains constant.

EXAMPLE 18.3 Mass of air in a scuba tank

WITH VARIATION PROBLEMS

An “empty” aluminum scuba tank contains 11.0 L of air at 21°C and 1 atm. When the tank is filled rapidly from a compressor, the air temperature is 42°C and the gauge pressure is $2.10 \times 10^7 \text{ Pa}$. What mass of air was added? (Air is about 78% nitrogen, 21% oxygen, and 1% miscellaneous; its average molar mass is $28.8 \text{ g/mol} = 28.8 \times 10^{-3} \text{ kg/mol}$.)

IDENTIFY and SET UP Our target variable is the difference $m_2 - m_1$ between the masses present at the end (state 2) and at the beginning (state 1). We are given the molar mass M of air, so we can use Eq. (18.2) to find the target variable if we know the number of moles present in states 1 and 2. We determine n_1 and n_2 by applying Eq. (18.3) to each state individually.

EXECUTE We convert temperatures to the Kelvin scale by adding 273 and convert the pressure to absolute by adding $1.013 \times 10^5 \text{ Pa}$. The tank’s volume is hardly affected by the increased temperature and pressure, so $V_2 = V_1$. From Eq. (18.3), the numbers of moles in the empty tank (n_1) and the full tank (n_2) are as follows:

$$n_1 = \frac{p_1 V_1}{RT_1} = \frac{(1.013 \times 10^5 \text{ Pa})(11.0 \times 10^{-3} \text{ m}^3)}{(8.314 \text{ J/mol} \cdot \text{K})(294 \text{ K})} = 0.46 \text{ mol}$$

$$n_2 = \frac{p_2 V_2}{RT_2} = \frac{(2.11 \times 10^7 \text{ Pa})(11.0 \times 10^{-3} \text{ m}^3)}{(8.314 \text{ J/mol} \cdot \text{K})(315 \text{ K})} = 88.6 \text{ mol}$$

We added $n_2 - n_1 = 88.6 \text{ mol} - 0.46 \text{ mol} = 88.1 \text{ mol}$ to the tank. From Eq. (18.2), the added mass is $M(n_2 - n_1) = (28.8 \times 10^{-3} \text{ kg/mol})(88.1 \text{ mol}) = 2.54 \text{ kg}$.

EVALUATE The added mass is not insubstantial: You could certainly use a scale to determine whether the tank was empty or full.

KEYCONCEPT You can determine the mass of a quantity of an ideal gas from its pressure, volume, and absolute temperature, and the molar mass of the gas.

EXAMPLE 18.4 Variation of atmospheric pressure with elevation

WITH VARIATION PROBLEMS

Find the variation of atmospheric pressure with elevation in the earth’s atmosphere. Assume that at all elevations, $T = 0^\circ\text{C}$ and $g = 9.80 \text{ m/s}^2$.

IDENTIFY and SET UP As the elevation y increases, both the atmospheric pressure p and the density ρ decrease. Hence we have *two* unknown functions of y ; to solve for them, we need two independent equations. One is the ideal-gas equation, Eq. (18.5), which is expressed in terms of p and ρ . The other is Eq. (12.4), the relationship that we found in Section 12.2 among p , ρ , and y in a fluid in equilibrium: $dp/dy = -\rho g$. We are told to assume that g and T are the same at all elevations; we also assume that the atmosphere has the same chemical composition, and hence the same molar mass M , at all heights. We combine the two equations and solve for $p(y)$.

EXECUTE We substitute $\rho = pM/RT$ into $dp/dy = -\rho g$, separate variables, and integrate, letting p_1 be the pressure at elevation y_1 and p_2 be the pressure at y_2 :

$$\frac{dp}{dy} = -\frac{pM}{RT}g$$

$$\int_{p_1}^{p_2} \frac{dp}{p} = -\frac{Mg}{RT} \int_{y_1}^{y_2} dy$$

$$\ln \frac{p_2}{p_1} = -\frac{Mg}{RT} (y_2 - y_1)$$

$$\frac{p_2}{p_1} = e^{-Mg(y_2 - y_1)/RT}$$

Now let $y_1 = 0$ be at sea level and let the pressure at that point be $p_0 = 1.013 \times 10^5 \text{ Pa}$. Then the pressure p at any height y is

$$p = p_0 e^{-Mgy/RT}$$

Continued

EVALUATE According to our calculation, the pressure decreases exponentially with elevation. The graph in **Fig. 18.4** shows that the slope dp/dy becomes less negative with greater elevation. That result makes sense, since $dp/dy = -\rho g$ and the density also decreases with elevation. At the summit of Mount Everest, where $y = 8848$ m,

$$\frac{Mgy}{RT} = \frac{(28.8 \times 10^{-3} \text{ kg/mol})(9.80 \text{ m/s}^2)(8848 \text{ m})}{(8.314 \text{ J/mol} \cdot \text{K})(273 \text{ K})} = 1.10$$

$$p = (1.013 \times 10^5 \text{ Pa})e^{-1.10} = 0.337 \times 10^5 \text{ Pa} = 0.33 \text{ atm}$$

The assumption of constant temperature isn't realistic, and g decreases a little with increasing elevation (see Challenge Problem 18.84). Even so, this example shows why most mountaineers carry oxygen on Mount Everest. It also shows why jet airliners, which typically fly at altitudes of 8000 to 12,000 m, *must* have pressurized cabins for passenger comfort and health.

KEYCONCEPT The ideal-gas equation can also be expressed as a relationship among the pressure, density, molar mass, and absolute temperature of an ideal gas [Eq. (12.4)].

Figure 18.4 The variation of atmospheric pressure p with elevation y , assuming a constant temperature T .

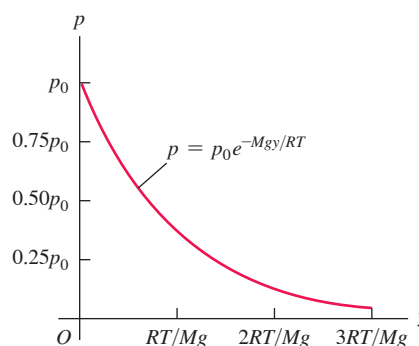
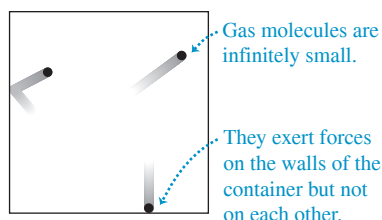
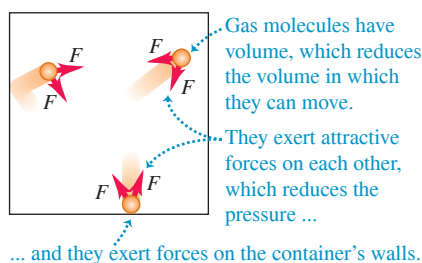


Figure 18.5 A gas as modeled by (a) the ideal-gas equation and (b) the van der Waals equation.

(a) An idealized model of a gas



(b) A more realistic model of a gas



The van der Waals Equation

In Section 18.3 we'll obtain the ideal-gas equation, Eq. (18.3), from a simple molecular model that ignores the volumes of the molecules themselves and the attractive forces between them (**Fig. 18.5a**). Another equation of state, the **van der Waals equation**, makes approximate corrections for these two omissions (**Fig. 18.5b**). This equation was developed by the 19th-century Dutch physicist J. D. van der Waals; the interaction between atoms that we discussed in Section 14.4 is named the *van der Waals interaction*. The van der Waals equation is

$$\left(p + \frac{an^2}{V^2}\right)(V - nb) = nRT \quad (18.7)$$

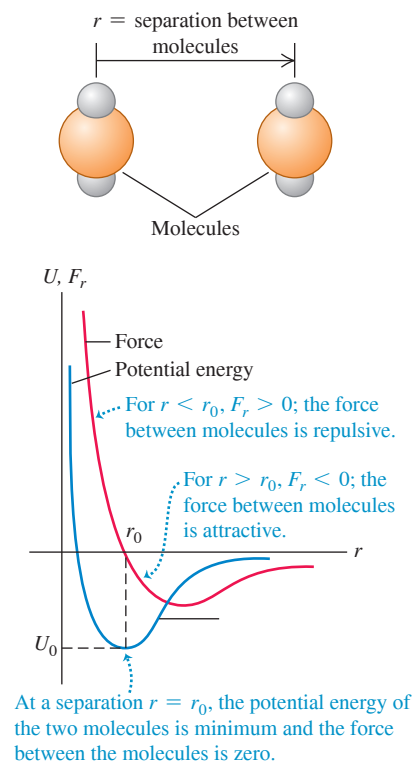
The constants a and b are different for different gases. Roughly speaking, b represents the volume of a mole of molecules; the total volume of the molecules is nb , and the volume remaining in which the molecules can move is $V - nb$. The constant a depends on the attractive intermolecular forces, which reduce the pressure of the gas by *pulling* the molecules together as they *push* on the walls of the container. The decrease in pressure is proportional to the number of molecules per unit volume in a layer near the wall (which are exerting the pressure on the wall) and is also proportional to the number per unit volume in the next layer beyond the wall (which are doing the attracting). Hence the decrease in pressure due to intermolecular forces is proportional to n^2/V^2 .

When n/V is small (that is, when the gas is *dilute*), the average distance between molecules is large, the corrections in the van der Waals equation become insignificant, and Eq. (18.7) reduces to the ideal-gas equation. As an example, for carbon dioxide gas (CO_2) the constants in the van der Waals equation are $a = 0.364 \text{ J} \cdot \text{m}^3/\text{mol}^2$ and $b = 4.27 \times 10^{-5} \text{ m}^3/\text{mol}$. We saw in Example 18.1 that 1 mole of an ideal gas at $T = 0^\circ\text{C} = 273.15 \text{ K}$ and $p = 1 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$ occupies a volume $V = 0.0224 \text{ m}^3$; according to Eq. (18.7), 1 mole of CO_2 occupying this volume at this temperature would be at a pressure 532 Pa less than 1 atm, a difference of only 0.5% from the ideal-gas value.

pV -Diagrams

We could in principle represent the p - V - T relationship graphically as a *surface* in a three-dimensional space with coordinates p , V , and T . This representation is useful (see Section 18.6), but ordinary two-dimensional graphs are usually more convenient. One of the most useful of these is a set of graphs of pressure as a function of volume, each for a particular constant temperature. Such a diagram is called a **pV -diagram**. Each curve, representing behavior at a specific temperature, is called an **isotherm**, or a **pV -isotherm**.

Figure 18.8 How the force between molecules and their potential energy of interaction depend on their separation r .



The interaction of two *point* electric charges is described by a force (repulsive for like charges, attractive for unlike charges) with a magnitude proportional to $1/r^2$, where r is the distance between the points. We'll study this relationship, called *Coulomb's law*, in Chapter 21. Molecules are *not* point charges but complex structures containing both positive and negative charge, and their interactions are more complex. The force between molecules in a gas varies with the distance r between molecules somewhat as shown in **Fig. 18.8**, where a positive F_r corresponds to a repulsive force and a negative F_r to an attractive force. When molecules are far apart, the intermolecular forces are very small and usually attractive. As a gas is compressed and its molecules are brought closer together, the attractive forces increase. The intermolecular force becomes zero at an equilibrium spacing r_0 , corresponding roughly to the spacing between molecules in the liquid and solid states. In liquids and solids, relatively large pressures are needed to compress the substance appreciably. This shows that at molecular distances slightly *less* than r_0 , the forces become *repulsive* and relatively large.

Figure 18.8 also shows the potential energy as a function of r . This function has a *minimum* at r_0 , where the force is zero. The two curves are related by $F_r(r) = -dU/dr$, as we showed in Section 7.4. Such a potential-energy function is often called a **potential well**. A molecule at rest at a distance r_0 from a second molecule would need an additional energy $|U_0|$, the "depth" of the potential well, to "escape" to an indefinitely large value of r .

Molecules are always in motion; their kinetic energies usually increase with temperature. At very low temperatures the average kinetic energy of a molecule may be much *less* than the depth of the potential well. The molecules then condense into the liquid or solid phase with average intermolecular spacings of about r_0 . But at higher temperatures the average kinetic energy becomes larger than the depth $|U_0|$ of the potential well. Molecules can then escape the intermolecular force and become free to move independently, as in the gaseous phase of matter.

In *solids*, molecules vibrate about more or less fixed points. (See Section 17.4.) In a crystalline solid these points are arranged in a *crystal lattice*. **Figure 18.9** shows the cubic crystal structure of sodium chloride, and **Fig. 18.10** shows a scanning tunneling microscope image of individual silicon atoms on the surface of a crystal.

In a *liquid*, the intermolecular distances are usually only slightly greater than in the solid phase of the same substance, but the molecules have much greater freedom of movement. Liquids show regularity of structure only in the immediate neighborhood of a few molecules.

Figure 18.9 Schematic representation of the cubic crystal structure of sodium chloride (ordinary salt).

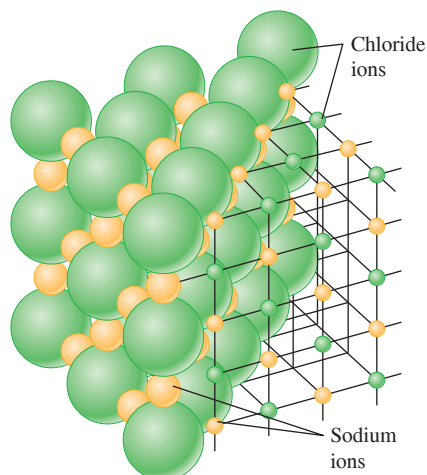
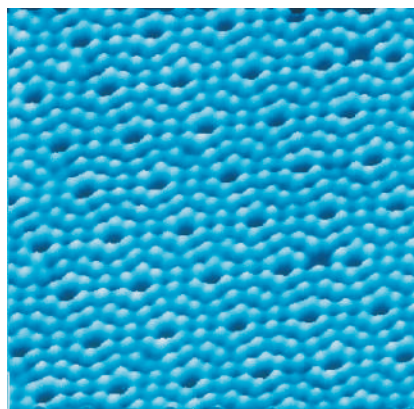


Figure 18.10 A scanning tunneling microscope image of the surface of a silicon crystal. The area shown is only 9.0 nm (9.0×10^{-9} m) across. Each blue "bead" is one silicon atom; these atoms are arranged in a (nearly) perfect array of hexagons.



The molecules of a *gas* are usually widely separated and so have only very small attractive forces. A gas molecule moves in a straight line until it collides with another molecule or with a wall of the container. In molecular terms, an *ideal gas* is a gas whose molecules exert *no* attractive forces on each other (see Fig. 18.5a) and therefore have no *potential* energy.

At low temperatures, most common substances are in the solid phase. As the temperature rises, a substance melts and then vaporizes. From a molecular point of view, these transitions are in the direction of increasing molecular kinetic energy. Thus temperature and molecular kinetic energy are closely related.

Moles and Avogadro's Number

We have used the mole as a measure of quantity of substance. One **mole** of any pure chemical element or compound contains a definite number of molecules, the same number for all elements and compounds. The official SI definition is:

One mole is the amount of substance that contains $6.02214076 \times 10^{23}$ elementary entities.

In our discussion, the “elementary entities” are molecules. (In a monatomic substance such as carbon or helium, each molecule is a single atom.) Until 2018 the number of elementary entities per mole was defined in terms of the number of atoms in 0.012 kilogram of carbon-12. The new definition is based on the redefinition of the kilogram (see Section 1.3) and on careful measurements of nearly perfect spheres of pure silicon-28.

The number of molecules in a mole is called **Avogadro's number**, denoted by N_A . The numerical value of N_A is *defined* to be

$$N_A = 6.02214076 \times 10^{23} \text{ molecules/mol} \quad (\text{Avogadro's number})$$

The *molar mass* M of a compound is the mass of 1 mole. It is equal to the mass m of a single molecule multiplied by Avogadro's number:

$$\text{Molar mass of a substance} \quad M = N_A \text{Mass of a molecule of substance} \quad (18.8)$$

When the molecule consists of a single atom, the term *atomic mass* is often used instead of molar mass.

EXAMPLE 18.5 Atomic and molecular mass

Find the mass of a single hydrogen atom and of a single oxygen molecule.

IDENTIFY and SET UP This problem involves the relationship between the mass of a molecule or atom (our target variable) and the corresponding molar mass M . We use Eq. (18.8) in the form $m = M/N_A$ and the values of the atomic masses from the periodic table of the elements (see Appendix D).

EXECUTE For atomic hydrogen the atomic mass (molar mass) is $M_H = 1.008 \text{ g/mol}$, so the mass m_H of a single hydrogen atom is

$$m_H = \frac{1.008 \text{ g/mol}}{6.022 \times 10^{23} \text{ atoms/mol}} = 1.674 \times 10^{-24} \text{ g/atom}$$

For oxygen the atomic mass is 16.0 g/mol , so for the diatomic (two-atom) oxygen molecule the molar mass is 32.0 g/mol . Then the mass of a single oxygen molecule is

$$m_{O_2} = \frac{32.0 \text{ g/mol}}{6.022 \times 10^{23} \text{ molecules/mol}} = 53.1 \times 10^{-24} \text{ g/molecule}$$

EVALUATE We note that the values in Appendix D are for the *average* atomic masses of a natural sample of each element. Such a sample may contain several *isotopes* of the element, each with a different atomic mass. Natural samples of hydrogen and oxygen are almost entirely made up of just one isotope.

KEYCONCEPT To find the mass of a single molecule of a substance, divide the molar mass of that substance by Avogadro's number (the number of molecules in a mole).

TEST YOUR UNDERSTANDING OF SECTION 18.2 Suppose you could adjust the value of r_0 for the molecules of a certain chemical compound (Fig. 18.8) by turning a dial. If you doubled the value of r_0 , the density of the solid form of this compound would become (i) twice as great; (ii) four times as great; (iii) eight times as great; (iv) $\frac{1}{2}$ as great; (v) $\frac{1}{4}$ as great; (vi) $\frac{1}{8}$ as great.

ANSWER (vi) The value of r_0 determines the equilibrium separation of the molecules in the solid phase, so doubling r_0 means that the separation doubles as well. Hence a solid cube of this compound might grow from 1 cm on a side to 2 cm on a side. The volume would then be $2^3 = 8$ times larger, and the density (mass divided by volume) would be $\frac{1}{8}$ as great.

18.3 KINETIC-MOLECULAR MODEL OF AN IDEAL GAS

The goal of any molecular theory of matter is to understand the *macroscopic* properties of matter in terms of its atomic or molecular structure and behavior. Once we have this understanding, we can design materials to have specific desired properties. Theories have led to the development of high-strength steels, semiconductor materials for electronic devices, and countless other materials essential to contemporary technology.

Let's consider a simple molecular model of an ideal gas. This *kinetic-molecular model* represents the gas as a large number of particles bouncing around in a closed container. In this section we use the kinetic-molecular model to understand how the ideal-gas equation of state, Eq. (18.3), is related to Newton's laws. In the following section we use the kinetic-molecular model to predict the molar heat capacity of an ideal gas. We'll go on to elaborate the model to include "particles" that are not points but have a finite size.

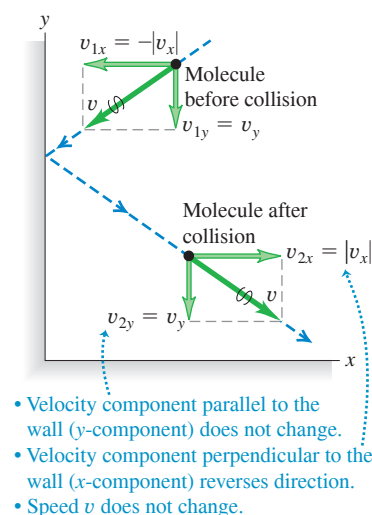
Our discussion of the kinetic-molecular model has several steps, and you may need to go over them several times. Don't get discouraged!

Here are the assumptions of our model:

1. A container with volume V contains a very large number N of identical molecules, each with mass m .
2. The molecules behave as point particles that are small compared to the size of the container and to the average distance between molecules.
3. The molecules are in constant motion. Each molecule collides occasionally with a wall of the container. These collisions are perfectly elastic.
4. The container walls are rigid and infinitely massive and do not move.

CAUTION Molecules vs. moles Don't confuse N , the number of *molecules* in the gas, with n , the number of *moles*. The number of molecules is equal to the number of moles multiplied by Avogadro's number: $N = nN_A$.

Figure 18.11 Elastic collision of a molecule with an idealized container wall.



Collisions and Gas Pressure

During collisions the molecules exert *forces* on the walls of the container; this is the origin of the *pressure* that the gas exerts. In a typical collision (Fig. 18.11) the velocity component parallel to the wall is unchanged, and the component perpendicular to the wall reverses direction but does not change in magnitude.

We'll first determine the *number* of collisions that occur per unit time for a certain area A of wall. Then we find the total momentum change associated with these collisions and the force needed to cause this momentum change. From this we can determine the pressure (force per unit area) and compare to the ideal-gas equation. We'll find a direct connection between the temperature of the gas and the kinetic energy of its molecules.

To begin, we'll assume that all molecules in the gas have the same *magnitude* of x -velocity, $|v_x|$. Later we'll see that our results don't depend on making this overly simplistic assumption.

As Fig. 18.11 shows, for each collision the x -component of velocity changes from $-|v_x|$ to $+|v_x|$. So the x -component of momentum p_x changes from $-m|v_x|$ to $+m|v_x|$, and the change in p_x is $m|v_x| - (-m|v_x|) = 2m|v_x|$.

If a molecule is going to collide with a given wall area A during a small time interval dt , then at the beginning of dt it must be within a distance $|v_x| dt$ from the wall (**Fig. 18.12**) and it must be headed toward the wall. So the number of molecules that collide with A during dt is equal to the number of molecules within a cylinder with base area A and length $|v_x| dt$ that have their x -velocity aimed toward the wall. The volume of such a cylinder is $A|v_x| dt$. Assuming that the number of molecules per unit volume (N/V) is uniform, the number of molecules in this cylinder is $(N/V)(A|v_x| dt)$. On the average, half of these molecules are moving toward the wall and half are moving away from it. So the number of collisions with A during dt is

$$\frac{1}{2} \left(\frac{N}{V} \right) (A|v_x| dt)$$

For the system of all molecules in the gas, the total momentum change dP_x during dt is the number of collisions multiplied by $2m|v_x|$:

$$dP_x = \frac{1}{2} \left(\frac{N}{V} \right) (A|v_x| dt) (2m|v_x|) = \frac{NAmv_x^2 dt}{V} \quad (18.9)$$

(We are using capital P for total momentum and lowercase p for pressure. Be careful!) We wrote v_x^2 rather than $|v_x|^2$ in the final expression because the square of the absolute value of a number is equal to the square of that number. The rate of change of momentum component P_x is

$$\frac{dP_x}{dt} = \frac{NAmv_x^2}{V} \quad (18.10)$$

According to Newton's second law, this rate of change of momentum equals the force exerted by the wall area A on the gas molecules. From Newton's *third* law this is equal and opposite to the force exerted on the wall by the molecules. Pressure p is the magnitude of the force exerted on the wall per unit area:

$$p = \frac{F}{A} = \frac{Nmv_x^2}{V} \quad (18.11)$$

The pressure exerted by the gas depends on the number of molecules per volume (N/V), the mass m per molecule, and the speed of the molecules.

Pressure and Molecular Kinetic Energies

We mentioned that $|v_x|$ is really *not* the same for all the molecules. But we could have sorted the molecules into groups having the same $|v_x|$ within each group, then added up the resulting contributions to the pressure. The net effect of all this is just to replace v_x^2 in Eq. (18.11) by the average value of v_x^2 , which we denote by $(v_x^2)_{av}$. We can relate $(v_x^2)_{av}$ to the speeds of the molecules. The speed v of a molecule is related to the velocity components v_x , v_y , and v_z by

$$v^2 = v_x^2 + v_y^2 + v_z^2$$

We can average this relationship over all molecules:

$$(v^2)_{av} = (v_x^2)_{av} + (v_y^2)_{av} + (v_z^2)_{av}$$

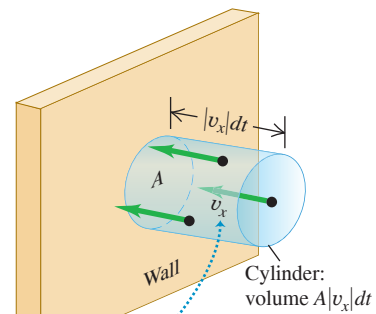
But there is no real difference in our model between the x -, y -, and z -directions. (Molecular speeds are very fast in a typical gas, so the effects of gravity are negligibly small.) It follows that $(v_x^2)_{av}$, $(v_y^2)_{av}$, and $(v_z^2)_{av}$ must all be *equal*. Hence $(v^2)_{av}$ is equal to $3(v_x^2)_{av}$ and

$$(v_x^2)_{av} = \frac{1}{3}(v^2)_{av}$$

so Eq. (18.11) becomes

$$pV = \frac{1}{3}Nm(v^2)_{av} = \frac{2}{3}N \left[\frac{1}{2}m(v^2)_{av} \right] \quad (18.12)$$

Figure 18.12 For a molecule to strike the wall in area A during a time interval dt , the molecule must be headed for the wall and be within the shaded cylinder of length $|v_x| dt$ at the beginning of the interval.



All molecules are assumed to have the same magnitude $|v_x|$ of x -velocity.

Figure 18.13 Summer air (top) is warmer than winter air (bottom); that is, the average translational kinetic energy of air molecules is greater in summer.



We notice that $\frac{1}{2}m(v^2)_{\text{av}}$ is the average translational kinetic energy of a single molecule. The product of this and the total number of molecules N equals the total random kinetic energy K_{tr} of translational motion of all the molecules. (The notation K_{tr} reminds us that this is the energy of *translational* motion. There may also be energies associated with molecular rotation and vibration.) The product pV equals two-thirds of the total translational kinetic energy:

$$pV = \frac{2}{3}K_{\text{tr}} \quad (18.13)$$

Now compare Eq. (18.13) to the ideal-gas equation $pV = nRT$, Eq. (18.3), which is based on experimental studies of gas behavior. For the two equations to agree, we must have

$$\text{Average translational kinetic energy of an ideal gas} \rightarrow K_{\text{tr}} = \frac{3}{2}nRT \leftarrow \begin{array}{l} \text{Number of moles of gas} \\ \text{Absolute temperature of gas} \\ \text{Gas constant} \end{array} \quad (18.14)$$

So K_{tr} is *directly proportional* to the absolute temperature T (**Fig. 18.13**).

The average translational kinetic energy of a single molecule is the total translational kinetic energy K_{tr} of all molecules divided by the number of molecules, N :

$$\frac{K_{\text{tr}}}{N} = \frac{1}{2}m(v^2)_{\text{av}} = \frac{3nRT}{2N}$$

Also, the total number of molecules N is the number of moles n multiplied by Avogadro's number N_A , so $N = nN_A$ and $n/N = 1/N_A$. Thus the above equation becomes

$$\frac{K_{\text{tr}}}{N} = \frac{1}{2}m(v^2)_{\text{av}} = \frac{3}{2}\left(\frac{R}{N_A}\right)T \quad (18.15)$$

The ratio R/N_A is called the **Boltzmann constant**, k :

$$k = \frac{R}{N_A} = \frac{8.314 \text{ J/mol} \cdot \text{K}}{6.022 \times 10^{23} \text{ molecules/mol}} = 1.381 \times 10^{-23} \text{ J/molecule} \cdot \text{K}$$

(The exact numerical value of k is *defined* to be $1.380649 \times 10^{-23} \text{ J/molecule} \cdot \text{K}$.) In terms of k we can rewrite Eq. (18.15) as

$$\text{Average translational kinetic energy of a gas molecule:} \rightarrow \frac{1}{2}m(v^2)_{\text{av}} = \frac{3}{2}kT \leftarrow \begin{array}{l} \text{Mass of a molecule} \\ \text{Absolute temperature of gas} \\ \text{Boltzmann constant} \end{array} \quad (18.16)$$

This shows that the average translational kinetic energy *per molecule* depends only on the temperature, not on the pressure, volume, or kind of molecule. We can obtain the average translational kinetic energy *per mole* by multiplying Eq. (18.16) by Avogadro's number and using the relationship $M = N_A m$:

$$N_A \frac{1}{2}m(v^2)_{\text{av}} = \frac{1}{2}M(v^2)_{\text{av}} = \frac{3}{2}RT \quad \text{(average translational kinetic energy per mole of gas)} \quad (18.17)$$

The translational kinetic energy of a mole of an ideal gas depends only on T .

Finally, it can be helpful to rewrite the ideal-gas equation on a “per-molecule” basis. We use $N = N_A n$ and $R = N_A k$ to obtain this alternative form:

$$pV = NkT \quad (18.18)$$

This shows that we can think of the Boltzmann constant k as a gas constant on a “per-molecule” basis instead of the usual “per-mole” basis for R .

CAUTION The Boltzmann constant, Avogadro's number, the gas constant, and the Kelvin scale In our discussion we began with the gas constant R and Avogadro's number N_A and used them to define the Boltzmann constant k . In fact, in the International System (SI; see Section 1.3), both N_A and k are fundamental quantities with defined values. The value of the gas constant is the product of these two defined values: $R = N_A k$. Furthermore, the value of k provides the official definition of the Kelvin scale: If the temperature increases by one kelvin, the quantity kT increases by exactly $1.380649 \times 10^{-23} \text{ J}$.

Molecular Speeds

From Eqs. (18.16) and (18.17) we can obtain expressions for the square root of $(v^2)_{\text{av}}$, ?
called the **root-mean-square speed** (or **rms speed**) v_{rms} :

$$v_{\text{rms}} = \sqrt{(v^2)_{\text{av}}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}} \quad (18.19)$$

Diagram labels for Equation 18.19:

- Root-mean-square speed of a gas molecule** points to v_{rms} .
- Average value of the square of molecular speeds** points to $(v^2)_{\text{av}}$.
- Boltzmann constant** points to k .
- Absolute temperature of gas** points to T .
- Mass of a molecule** points to m .
- Gas constant** points to R .
- Molar mass** points to M .

It might seem more natural to give the *average* speed rather than v_{rms} , but v_{rms} follows more directly from Eqs. (18.16) and (18.17). To compute the rms speed, we square each molecular speed, add, divide by the number of molecules, and take the square root; v_{rms} is the *root of the mean of the squares*.

Equations (18.16) and (18.19) show that at a given temperature T , gas molecules of different mass m have the same average kinetic energy but different root-mean-square speeds. On average, the nitrogen molecules ($M = 28 \text{ g/mol}$) in the air around you are moving faster than are the oxygen molecules ($M = 32 \text{ g/mol}$). Hydrogen molecules ($M = 2 \text{ g/mol}$) are fastest of all; this is why there is hardly any hydrogen in the earth's atmosphere, despite its being the most common element in the universe (**Fig. 18.14**). A sizable fraction of any H_2 molecules in the atmosphere would have speeds greater than the earth's escape speed of $1.12 \times 10^4 \text{ m/s}$ (calculated in Example 13.5 in Section 13.3) and would escape into space. The heavier, slower-moving gases cannot escape so easily, which is why they predominate in our atmosphere.

The assumption that individual molecules undergo perfectly elastic collisions with the container wall is a little too simple. In most cases, molecules actually adhere to the wall for a short time and then leave again with speeds that are characteristic of the temperature *of the wall*. However, the gas and the wall are ordinarily in thermal equilibrium and have the same temperature. So there is no net energy transfer between gas and wall, and our conclusions remain valid.

Figure 18.14 While hydrogen is a desirable fuel for vehicles, it is only a trace constituent of our atmosphere (0.00005% by volume). Hence hydrogen fuel has to be generated by electrolysis of water, which is itself an energy-intensive process.



PROBLEM-SOLVING STRATEGY 18.2 Kinetic-Molecular Theory

IDENTIFY the relevant concepts: Use the results of the kinetic-molecular model to relate the macroscopic properties of a gas, such as temperature and pressure, to microscopic properties, such as molecular speeds.

SET UP the problem using the following steps:

1. List knowns and unknowns; identify the target variables.
2. Choose appropriate equation(s) from among Eqs. (18.14), (18.16), and (18.19).

EXECUTE the solution as follows: Maintain consistency in units.

1. The usual units for molar mass M are grams per mole; these units are often omitted in tables. In equations such as Eq. (18.19), when you use SI units you must express M in kilograms per mole. For example, for oxygen $M_{\text{O}_2} = 32 \text{ g/mol} = 32 \times 10^{-3} \text{ kg/mol}$.

2. Are you working on a “per-molecule” basis (with m , N , and k) or a “per-mole” basis (with M , n , and R)? To check units, think of N as having units of “molecules”; then m has units of mass per molecule, and k has units of joules per molecule per kelvin. Similarly, n has units of moles; then M has units of mass per mole and R has units of joules per mole per kelvin.
3. Remember that T is always *absolute* (Kelvin) temperature.

EVALUATE your answer: Are your answers reasonable? Here's a benchmark: Typical molecular speeds at room temperature are several hundred meters per second.

EXAMPLE 18.6 Molecular kinetic energy and v_{rms} **WITH VARIATION PROBLEMS**

(a) What is the average translational kinetic energy of an ideal-gas molecule at 27°C ? (b) What is the total random translational kinetic energy of the molecules in 1 mole of this gas? (c) What is the rms speed of oxygen molecules at this temperature?

IDENTIFY and SET UP This problem involves the translational kinetic energy of an ideal gas on a per-molecule and per-mole basis, as well as the root-mean-square molecular speed v_{rms} . We are given $T = 27^\circ\text{C} = 300\text{ K}$ and $n = 1\text{ mol}$; we use the molecular mass m for oxygen. We use Eq. (18.16) to determine the average kinetic energy of a molecule, Eq. (18.14) to find the total molecular kinetic energy K_{tr} of 1 mole, and Eq. (18.19) to find v_{rms} .

EXECUTE (a) From Eq. (18.16),

$$\frac{1}{2}m(v^2)_{\text{av}} = \frac{3}{2}kT = \frac{3}{2}(1.38 \times 10^{-23}\text{ J/K})(300\text{ K}) \\ = 6.21 \times 10^{-21}\text{ J}$$

(b) From Eq. (18.14), the kinetic energy of one mole is

$$K_{\text{tr}} = \frac{3}{2}nRT = \frac{3}{2}(1\text{ mol})(8.314\text{ J/mol} \cdot \text{K})(300\text{ K}) = 3740\text{ J}$$

(c) We found the mass per molecule m and molar mass M of molecular oxygen in Example 18.5. Using Eq. (18.19), we can calculate v_{rms} in two ways:

$$v_{\text{rms}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3(1.38 \times 10^{-23}\text{ J/K})(300\text{ K})}{5.31 \times 10^{-26}\text{ kg}}} \\ = 484\text{ m/s} = 1740\text{ km/h} = 1080\text{ mi/h} \\ v_{\text{rms}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3(8.314\text{ J/mol} \cdot \text{K})(300\text{ K})}{32.0 \times 10^{-3}\text{ kg/mol}}} = 484\text{ m/s}$$

EVALUATE The answer in part (a) does not depend on the mass of the molecule. We can check our result in part (b) by noting that the translational kinetic energy per mole must be equal to the product of the average translational kinetic energy per molecule from part (a) and Avogadro's number N_A : $K_{\text{tr}} = (6.21 \times 10^{-21}\text{ J/molecule}) \times (6.022 \times 10^{23}\text{ molecules}) = 3740\text{ J}$.

KEYCONCEPT The average translational kinetic energy of a molecule in an ideal gas at absolute temperature T is $\frac{3}{2}kT$, where k is the Boltzmann constant, no matter what the mass of the molecules. However, the root-mean-square speed v_{rms} of molecules (that is, the square root of the average value of v^2) in an ideal gas does depend on the mass per molecule m : $v_{\text{rms}} = \sqrt{\frac{3kT}{m}}$.

EXAMPLE 18.7 Calculating rms and average speeds**WITH VARIATION PROBLEMS**

Five gas molecules have speeds of 500, 600, 700, 800, and 900 m/s. What is the rms speed? What is the average speed?

IDENTIFY and SET UP We use the definitions of the root mean square and the average of a collection of numbers. To find v_{rms} , we square each speed, find the average (mean) of the squares, and take the square root of the result. We find v_{av} as usual.

EXECUTE The average value of v^2 and the resulting v_{rms} for the five molecules are

$$(v^2)_{\text{av}} = \frac{500^2 + 600^2 + 700^2 + 800^2 + 900^2}{5}\text{ m}^2/\text{s}^2 \\ = 5.10 \times 10^5\text{ m}^2/\text{s}^2 \\ v_{\text{rms}} = \sqrt{(v^2)_{\text{av}}} = 714\text{ m/s}$$

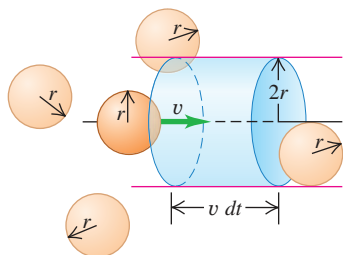
The average speed v_{av} is

$$v_{\text{av}} = \frac{500 + 600 + 700 + 800 + 900}{5}\text{ m/s} = 700\text{ m/s}$$

EVALUATE In general v_{rms} and v_{av} are *not* the same. Roughly speaking, v_{rms} gives greater weight to the higher speeds than does v_{av} .

KEYCONCEPT In general, for any collection of numbers, the root-mean-square value is *not* equal to the average value. For molecules in an ideal gas, the root-mean-square speed is always greater than the average speed.

Figure 18.15 In a time dt a molecule with radius r will collide with any other molecule within a cylindrical volume of radius $2r$ and length $v dt$.

**Collisions Between Molecules**

We have ignored the possibility that two gas molecules might collide. If they are really points, they *never* collide. But consider a more realistic model in which the molecules are rigid spheres with radius r . How often do they collide with other molecules? How far do they travel, on average, between collisions? We can get approximate answers from the following rather primitive model.

Consider N spherical molecules with radius r in a volume V . Suppose only one molecule is moving. When it collides with another molecule, the distance between centers is $2r$. Suppose we draw a cylinder with radius $2r$, with its axis parallel to the velocity of the molecule (**Fig. 18.15**). The moving molecule collides with any other molecule whose center is inside this cylinder. In a short time dt a molecule with speed v travels a distance $v dt$;

during this time it collides with any molecule that is in the cylindrical volume of radius $2r$ and length $v dt$. The volume of the cylinder is $4\pi r^2 v dt$. There are N/V molecules per unit volume, so the number dN with centers in this cylinder is

$$dN = 4\pi r^2 v dt N/V$$

Thus the number of collisions *per unit time* is

$$\frac{dN}{dt} = \frac{4\pi r^2 v N}{V}$$

This result assumes that only one molecule is moving. It turns out that collisions are more frequent when all the molecules move at once, and the above equation has to be multiplied by a factor of $\sqrt{2}$:

$$\frac{dN}{dt} = \frac{4\pi \sqrt{2} r^2 v N}{V}$$

The average time t_{mean} between collisions, called the *mean free time*, is the reciprocal of this expression:

$$t_{\text{mean}} = \frac{V}{4\pi \sqrt{2} r^2 v N} \quad (18.20)$$

The average distance traveled between collisions is called the **mean free path**. In our model, this is just the molecule's speed v multiplied by t_{mean} :

$$\lambda = v t_{\text{mean}} = \frac{V}{4\pi \sqrt{2} r^2 N} \quad (18.21)$$

Mean free path of a gas molecule λ is equal to Speed of molecule v multiplied by Mean free time between collisions t_{mean} . The volume of gas V is equal to the product of the Radius of a molecule r squared and the Number of molecules in gas N .

The mean free path λ (the Greek letter lambda) is inversely proportional to the number of molecules per unit volume (N/V) and inversely proportional to the cross-sectional area πr^2 of a molecule; the more molecules there are and the larger the molecule, the shorter the mean distance between collisions (**Fig. 18.16**). Note that the mean free path *does not* depend on the speed of the molecule.

We can express Eq. (18.21) in terms of macroscopic properties of the gas, using the ideal-gas equation in the form of Eq. (18.18), $pV = NkT$. We find

$$\lambda = \frac{kT}{4\pi \sqrt{2} r^2 p} \quad (18.22)$$

If the temperature is increased at constant pressure, the gas expands, the average distance between molecules increases, and λ increases. If the pressure is increased at constant temperature, the gas compresses and λ decreases.

Figure 18.16 If you try to walk through a crowd, your mean free path—the distance you can travel on average without running into another person—depends on how large the people are and how closely they are spaced.



EXAMPLE 18.8 Calculating mean free path

WITH VARIATION PROBLEMS

- Estimate the mean free path of a molecule of air at 27°C and 1 atm. Model the molecules as spheres with radius $r = 2.0 \times 10^{-10}$ m.
- Estimate the mean free time of an oxygen molecule with $v = v_{\text{rms}}$ at 27°C and 1 atm.

IDENTIFY and SET UP This problem uses the concepts of mean free path and mean free time (our target variables). We use Eq. (18.22) to

determine the mean free path λ . We then use $\lambda = v t_{\text{mean}}$ in Eq. (18.21), with $v = v_{\text{rms}}$, to find the mean free time t_{mean} .

EXECUTE (a) From Eq. (18.22),

$$\begin{aligned} \lambda &= \frac{kT}{4\pi \sqrt{2} r^2 p} = \frac{(1.38 \times 10^{-23} \text{ J/K})(300 \text{ K})}{4\pi \sqrt{2} (2.0 \times 10^{-10} \text{ m})^2 (1.01 \times 10^5 \text{ Pa})} \\ &= 5.8 \times 10^{-8} \text{ m} \end{aligned}$$

Continued

(b) From Example 18.6, for oxygen at 27°C the root-mean-square speed is $v_{\text{rms}} = 484 \text{ m/s}$, so the mean free time for a molecule with this speed is

$$t_{\text{mean}} = \frac{\lambda}{v} = \frac{5.8 \times 10^{-8} \text{ m}}{484 \text{ m/s}} = 1.2 \times 10^{-10} \text{ s}$$

This molecule undergoes about 10^{10} collisions per second!

EVALUATE Note that from Eqs. (18.21) and (18.22) the mean free path doesn't depend on the molecule's speed, but the mean free time does. Slower molecules have a longer average time interval t_{mean}

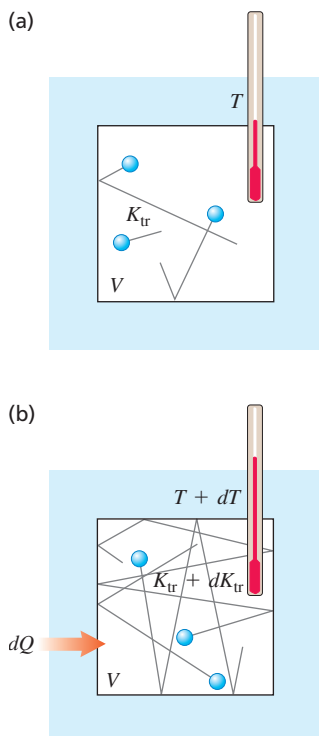
between collisions than do fast ones, but the average distance λ between collisions is the same no matter what the molecule's speed. Our answer to part (a) says that the molecule doesn't go far between collisions, but the mean free path is still several hundred times the molecular radius r .

KEYCONCEPT The mean free path and mean free time are, respectively, the average distance and average time that a molecule in a gas travels between collisions with other molecules. Both depend on the temperature and pressure of the gas and the radius of a molecule. The mean free time also depends on the speed of the molecule; the mean free path does not.

TEST YOUR UNDERSTANDING OF SECTION 18.3 Rank the following gases in order from (a) highest to lowest rms speed of molecules and (b) highest to lowest average translational kinetic energy of a molecule: (i) oxygen ($M = 32.0 \text{ g/mol}$) at 300 K; (ii) nitrogen ($M = 28.0 \text{ g/mol}$) at 300 K; (iii) oxygen at 330 K; (iv) nitrogen at 330 K.

ANSWER (a) (iv), (iii), (i), (ii); (b) (i), (ii), (iii), (iv). (a) Equation (18.19) tells us that $v_{\text{rms}} = \sqrt{3RT/M}$, so the rms speed is proportional to the square root of the ratio of absolute temperature T to molar mass M . Compared to (i) oxygen at 300 K, v_{rms} in the other cases is (ii) $\sqrt{(32.0 \text{ g/mol})/(28.0 \text{ g/mol})} = 1.07$ times faster, (iii) $\sqrt{(330 \text{ K})/(300 \text{ K})} = 1.05$ times faster, and (iv) $\sqrt{(330 \text{ K})/(28.0 \text{ g/mol})} = 1.12$ times faster. (b) From Eq. (18.16), the average translational kinetic energy per molecule is $\frac{1}{2}m(v_{\text{rms}}^2)_{\text{av}} = \frac{3}{2}kT$, which is directly proportional to T and independent of M . We have $T = 300 \text{ K}$ for cases (i) and (ii) and $T = 330 \text{ K}$ for cases (iii) and (iv), so $\frac{1}{2}m(v_{\text{rms}}^2)_{\text{av}}$ has equal values for cases (iii) and (iv) and equal (but smaller) values for cases (i) and (ii).

Figure 18.17 (a) A fixed volume V of a monatomic ideal gas. (b) When an amount of heat dQ is added to the gas, the total translational kinetic energy increases by $dK_{\text{tr}} = dQ$ and the temperature increases by $dT = dQ/nC_V$.



18.4 HEAT CAPACITIES

When we introduced the concept of heat capacity in Section 17.5, we talked about ways to *measure* the specific heat or molar heat capacity of a particular material. Now we'll see how to *predict* these on theoretical grounds.

Heat Capacities of Gases

The basis of our analysis is that heat is *energy* in transit. When we add heat to a substance, we are increasing its molecular energy. We'll assume that the volume of the gas remains constant; if we were to let the gas expand, it would do work by pushing on the moving walls of its container, and this additional energy transfer would have to be included in our calculations. We'll return to this more general case in Chapter 19. For now we are concerned with C_V , the molar heat capacity *at constant volume*.

In the simple kinetic-molecular model of Section 18.3 the molecular energy consists of only the translational kinetic energy K_{tr} of the pointlike molecules. This energy is directly proportional to the absolute temperature T , as shown by Eq. (18.14), $K_{\text{tr}} = \frac{3}{2}nRT$. When the temperature changes by a small amount dT , the corresponding change in kinetic energy is

$$dK_{\text{tr}} = \frac{3}{2}nR dT \quad (18.23)$$

From the definition of molar heat capacity at constant volume, C_V (see Section 17.5), we also have

$$dQ = nC_V dT \quad (18.24)$$

where dQ is the heat input needed for a temperature change dT . Now if K_{tr} represents the total molecular energy, as we have assumed, then dQ and dK_{tr} must be *equal* (**Fig. 18.17**). From Eqs. (18.23) and (18.24), this says

$$nC_V dT = \frac{3}{2}nR dT$$

Molar heat capacity at constant volume, ideal gas of point particles $\rightarrow C_V = \frac{3}{2}R \leftarrow$ Gas constant (18.25)

This surprisingly simple result says that the molar heat capacity at constant volume is $\frac{3}{2}R$ for *any* gas whose molecules can be represented as points.

Does Eq. (18.25) agree with experiment? In SI units, Eq. (18.25) gives

$$C_V = \frac{3}{2}(8.314 \text{ J/mol} \cdot \text{K}) = 12.47 \text{ J/mol} \cdot \text{K}$$

For comparison, **Table 18.1** gives measured values of C_V for several gases. We see that for *monatomic* gases our prediction is right on the money, but that it is way off for diatomic and polyatomic gases.

This comparison tells us that our point-molecule model is good enough for monatomic gases but that for diatomic and polyatomic molecules we need something more sophisticated. For example, we can picture a diatomic molecule as *two* point masses, like a little elastic dumbbell (see **Fig. 18.18**), with an interaction force between the atoms of the kind shown in Fig. 18.8. Such a molecule can have additional kinetic energy associated with *rotation* about axes through its center of mass. The atoms may also vibrate along the line joining them, with additional kinetic and potential energies.

When heat flows into a *monatomic* gas at constant volume, *all* of the added energy goes into an increase in random *translational* molecular kinetic energy. Equation (18.23) shows that this gives rise to an increase in temperature. But when the temperature is increased by the same amount in a *diatomic* or *polyatomic* gas, additional heat is needed to supply the increased rotational and vibrational energies. Thus polyatomic gases have *larger* molar heat capacities than monatomic gases, as Table 18.1 shows.

But how do we know how much energy is associated with each additional kind of motion of a complex molecule, compared to the translational kinetic energy? The new principle that we need is called the principle of **equipartition of energy**. It can be derived from sophisticated statistical-mechanics considerations; that derivation is beyond our scope, and we'll treat the principle as an axiom.

The principle of equipartition of energy states that each velocity component (either linear or angular) has, on average, an associated kinetic energy per molecule of $\frac{1}{2}kT$, or one-half the product of the Boltzmann constant and the absolute temperature. The number of velocity components needed to describe the motion of a molecule completely is called the number of **degrees of freedom**. For a monatomic gas, there are three degrees of freedom (for the velocity components v_x , v_y , and v_z); this gives a total average kinetic energy per molecule of $3(\frac{1}{2}kT)$, consistent with Eq. (18.16).

For a *diatomic* molecule there are two possible axes of rotation, perpendicular to each other and to the molecule's axis. (We don't include rotation about the molecule's own axis because in ordinary collisions there is no way for this rotational motion to change.) If we add two rotational degrees of freedom for a diatomic molecule, the average total kinetic energy per molecule is $\frac{5}{2}kT$ instead of $\frac{3}{2}kT$. The total kinetic energy of n moles is $K_{\text{total}} = nN_A(\frac{5}{2}kT) = \frac{5}{2}n(kN_A)T = \frac{5}{2}nRT$, and the molar heat capacity (at constant volume) is

Molar heat capacity at constant volume, ideal diatomic gas $\rightarrow C_V = \frac{5}{2}R \leftarrow$ Gas constant (18.26)

In SI units,

$$C_V = \frac{5}{2}(8.314 \text{ J/mol} \cdot \text{K}) = 20.79 \text{ J/mol} \cdot \text{K}$$

This value is close to the measured values for diatomic gases in Table 18.1.

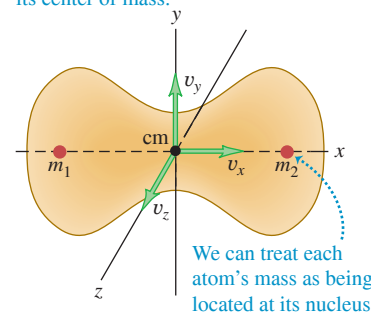
Vibrational motion can also contribute to the heat capacities of gases. Molecular bonds can stretch and bend, and the resulting vibrations lead to additional degrees of freedom and additional energies. For most diatomic gases, however, vibration does *not* contribute appreciably to heat capacity. The reason for this involves some concepts of quantum mechanics. Briefly, vibrational energy can change only in finite steps. If the energy change of the first step is much larger than the energy possessed by most molecules, then nearly all

TABLE 18.1 Molar Heat Capacities of Gases

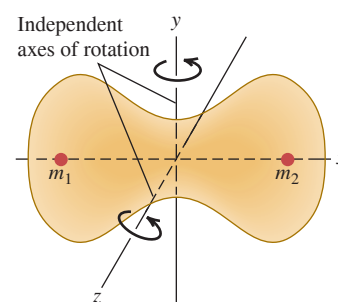
Type of Gas	Gas	C_V (J/mol · K)
Monatomic	He	12.47
	Ar	12.47
Diatomic	H ₂	20.42
	N ₂	20.76
	O ₂	20.85
	CO	20.85
Polyatomic	CO ₂	28.46
	SO ₂	31.39
	H ₂ S	25.95

Figure 18.18 Motions of a diatomic molecule.

(a) **Translational motion.** The molecule moves as a whole; its velocity may be described as the x -, y -, and z -velocity components of its center of mass.



(b) **Rotational motion.** The molecule rotates about its center of mass. This molecule has two independent axes of rotation.



(c) **Vibrational motion.** The molecule oscillates as though the nuclei were connected by a spring.

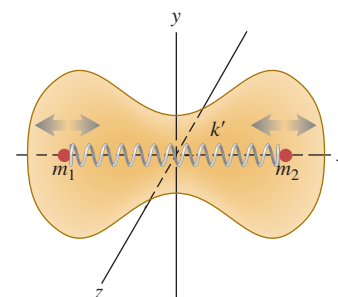
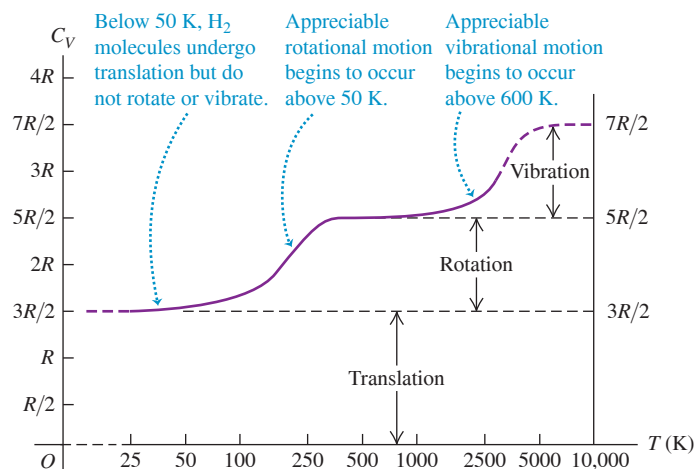


Figure 18.19 Experimental values of C_V , the molar heat capacity at constant volume, for hydrogen gas (H_2). The temperature is plotted on a logarithmic scale.



the molecules remain in the minimum-energy state of motion. Changing the temperature does not change their average vibrational energy appreciably, and the vibrational degrees of freedom are said to be “frozen out.” In more complex molecules the gaps between permitted energy levels can be much smaller, and then vibration *does* contribute to heat capacity. The rotational energy of a molecule also changes by finite steps, but they are usually much smaller; the “freezing out” of rotational degrees of freedom occurs only in rare instances.

In Table 18.1 the large values of C_V for polyatomic molecules show the effects of vibrational energy. In addition, a molecule with three or more atoms that are not in a straight line has *three* rotational degrees of freedom.

From this discussion we expect heat capacities to be temperature-dependent, generally increasing with increasing temperature. **Figure 18.19** is a graph of the temperature dependence of C_V for hydrogen gas (H_2), showing the temperatures at which the rotational and vibrational energies begin to contribute.

Heat Capacities of Solids

We can carry out a similar heat-capacity analysis for a crystalline solid. Consider a crystal consisting of N identical atoms (a *monatomic solid*). Each atom is bound to an equilibrium position by interatomic forces. Solid materials are elastic, so forces must permit stretching and bending of the bonds. We can think of a crystal as an array of atoms connected by little springs (**Fig. 18.20**).

Each atom can *vibrate* around its equilibrium position and has three degrees of freedom, corresponding to its three components of velocity. According to the equipartition principle, each atom has an average kinetic energy of $\frac{1}{2}kT$ for each degree of freedom. In addition, there is *potential* energy associated with the elastic deformation. For a simple harmonic oscillator (discussed in Chapter 14) it is not hard to show that the average kinetic energy is *equal* to the average potential energy. In our model of a crystal, each atom is a three-dimensional harmonic oscillator; it can be shown that the equality of average kinetic and potential energies also holds here, provided that the “spring” forces obey Hooke’s law.

Thus we expect each atom to have an average kinetic energy $\frac{3}{2}kT$ and an average potential energy $\frac{3}{2}kT$, or an average total energy $3kT$ per atom. If the crystal contains N atoms or n moles, its total energy is

$$E_{\text{total}} = 3NkT = 3nRT \quad (18.27)$$

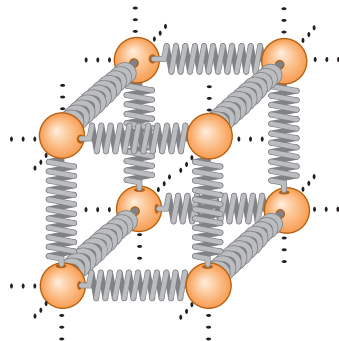
From this we conclude that the molar heat capacity of a crystal should be

$$\text{Molar heat capacity of an ideal monatomic solid (rule of Dulong and Petit)} \quad C_V = 3R \quad \text{Gas constant} \quad (18.28)$$

In SI units,

$$C_V = (3)(8.314 \text{ J/mol} \cdot \text{K}) = 24.9 \text{ J/mol} \cdot \text{K}$$

Figure 18.20 To visualize the forces between neighboring atoms in a crystal, envision every atom as being attached to its neighbors by springs.



We have *derived* the **rule of Dulong and Petit**, which we encountered as an *empirical* finding in Section 17.5: Monatomic solids all have molar heat capacities of about $25 \text{ J/mol} \cdot \text{K}$. The agreement is only approximate, but given the very simple nature of our model, it is quite significant.

At low temperatures, the heat capacities of most solids *decrease* with decreasing temperature (**Fig. 18.21**) for the same reason that vibrational degrees of freedom of molecules are frozen out at low temperatures. At very low temperatures the quantity kT is much *smaller* than the smallest energy step the vibrating atoms can take. Hence most of the atoms remain in their lowest energy states because the next higher energy level is out of reach. The average vibrational energy per atom is then *less* than $3kT$, and the heat capacity per molecule is *less* than $3k$. At higher temperatures when kT is *large* in comparison to the minimum energy step, the equipartition principle holds, and the total heat capacity is $3k$ per molecule or $3R$ per mole as the rule of Dulong and Petit predicts. Quantitative understanding of the temperature variation of heat capacities was one of the triumphs of quantum mechanics during its initial development in the 1920s.

TEST YOUR UNDERSTANDING OF SECTION 18.4 A cylinder with a fixed volume contains hydrogen gas (H_2) at 25 K. You then add heat to the gas at a constant rate until its temperature reaches 500 K. Does the temperature of the gas increase at a constant rate? Why or why not? If not, does the temperature increase most rapidly near the beginning or near the end of this process?

ANSWER

no; near the beginning Adding a small amount of heat dQ to the gas changes the temperature by dT , where $dQ = nC_V dT$ from Eq. (18.24). Figure 18.19 shows that C_V for H_2 varies with temperature between 25 K and 500 K, so a given amount of heat gives rise to different amounts of temperature change during the process. Hence the temperature will *not* increase at a constant rate. The temperature change $dT = dQ/nC_V$ is inversely proportional to C_V , so the temperature increases most rapidly at the beginning of the process when the temperature is lowest and C_V is smallest (see Fig. 18.19).

18.5 MOLECULAR SPEEDS

As we mentioned in Section 18.3, the molecules in a gas don't all have the same speed. **Figure 18.22** shows one experimental scheme for measuring the distribution of molecular speeds. A substance is vaporized in a hot oven; molecules of the vapor escape through an aperture in the oven wall and into a vacuum chamber. A series of slits blocks all molecules except those in a narrow beam, which is aimed at a pair of rotating disks. A molecule passing through the slit in the first disk is blocked by the second disk unless it arrives just as the slit in the second disk is lined up with the beam. The disks function as a speed selector that passes only molecules within a certain narrow speed range. This range can be varied by changing the disk rotation speed, and we can measure how many molecules lie within each of various speed ranges.

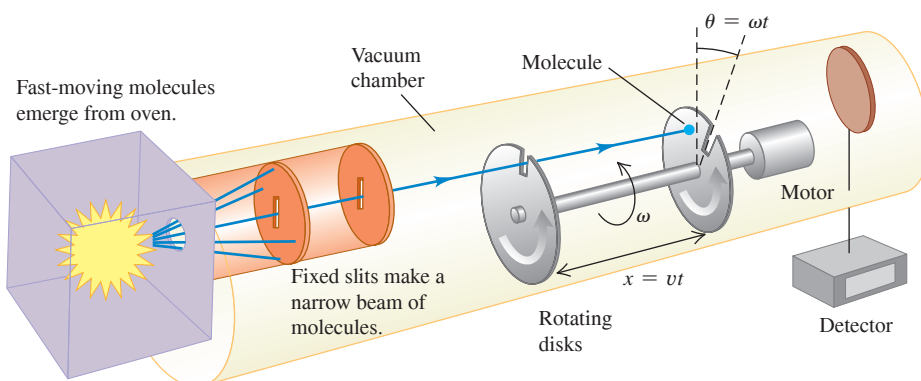


Figure 18.21 Experimental values of C_V for lead, aluminum, silicon, and diamond. At high temperatures, C_V for each solid approaches about $3R$, in agreement with the rule of Dulong and Petit. At low temperatures, C_V is much less than $3R$.

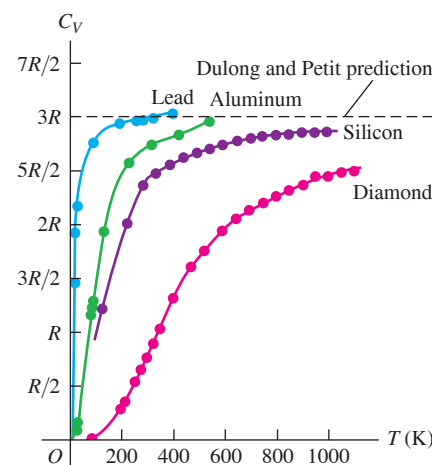
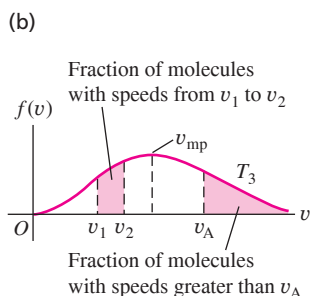
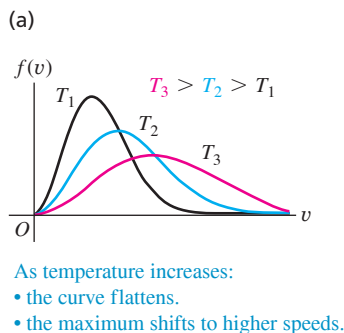


Figure 18.22 A molecule with a speed v passes through the slit in the first rotating disk. When the molecule reaches the second rotating disk, the disks have rotated through the offset angle θ . If $v = \omega x/\theta$, the molecule passes through the slit in the second rotating disk and reaches the detector.

To describe the results of such measurements, we define a function $f(v)$ called a *distribution function*. If we observe a total of N molecules, the number dN having speeds in the range between v and $v + dv$ is given by

$$dN = Nf(v) dv \quad (18.29)$$

Figure 18.23 (a) Curves of the Maxwell–Boltzmann distribution function $f(v)$ for three temperatures. (b) The shaded areas under the curve represent the fractions of molecules within certain speed ranges. The most probable speed v_{mp} for a given temperature is at the peak of the curve.



The *probability* that a randomly chosen molecule will have a speed in the interval v to $v + dv$ is $f(v)dv$. Hence $f(v)$ is the probability per unit speed *interval*; it is *not* the probability that a molecule has speed exactly equal to v . Since a probability is a pure number, $f(v)$ has units of reciprocal speed (s/m).

Figure 18.23a shows distribution functions for three different temperatures. At each temperature the height of the curve for any value of v is proportional to the number of molecules with speeds near v . The peak of the curve represents the *most probable speed* v_{mp} for the corresponding temperature. As the temperature increases, the average molecular kinetic energy increases, and so the peak of $f(v)$ shifts to higher and higher speeds.

Figure 18.23b shows that the area under a curve between any two values of v represents the fraction of all the molecules having speeds in that range. Every molecule must have *some* value of v , so the integral of $f(v)$ over all v must be unity for any T .

If we know $f(v)$, we can calculate the most probable speed v_{mp} , the average speed v_{av} , and the rms speed v_{rms} . To find v_{mp} , we simply find the point where $df/dv = 0$; this gives the value of the speed where the curve has its peak. To find v_{av} , we take the number $Nf(v) dv$ having speeds in each interval dv , multiply each number by the corresponding speed v , add all these products (by integrating over all v from zero to infinity), and finally divide by N . That is,

$$v_{\text{av}} = \int_0^\infty v f(v) dv \quad (18.30)$$

We can find the rms speed in a similar way; the average of v^2 is

$$(v^2)_{\text{av}} = \int_0^\infty v^2 f(v) dv \quad (18.31)$$

and v_{rms} is the square root of this.

The Maxwell–Boltzmann Distribution

The function $f(v)$ describing the actual distribution of molecular speeds is called the **Maxwell–Boltzmann distribution**. It can be derived from statistical-mechanics considerations, but that derivation is beyond our scope. Here is the result:

$$\text{Maxwell–Boltzmann distribution function } f(v) = 4\pi \left(\frac{m}{2\pi kT} \right)^{3/2} v^2 e^{-mv^2/2kT} \quad (18.32)$$

Diagram illustrating the components of the Maxwell-Boltzmann distribution function equation (18.32):

- Mass of a gas molecule** (m)
- Molecular speed** (v)
- Mass of a gas molecule** (m)
- Molecular speed** (v)
- Boltzmann constant** (k)
- Absolute temperature of gas** (T)

We can also express this function in terms of the translational kinetic energy of a molecule, which we denote by ϵ ; that is, $\epsilon = \frac{1}{2}mv^2$. We invite you to verify that when this is substituted into Eq. (18.32), the result is

$$f(\epsilon) = \frac{8\pi}{m} \left(\frac{m}{2\pi kT} \right)^{3/2} \epsilon e^{-\epsilon/kT} \quad (18.33)$$

This form shows that the exponent in the Maxwell–Boltzmann distribution function is $-\epsilon/kT$; the shape of the curve is determined by the relative magnitude of ϵ and kT at any point. You can prove that the *peak* of each curve occurs where $\epsilon = kT$, corresponding to a most probable speed v_{mp} :

$$v_{\text{mp}} = \sqrt{\frac{2kT}{m}} \quad (18.34)$$

To find the average speed, we substitute Eq. (18.32) into Eq. (18.30), make a change of variable $v^2 = x$, and integrate by parts. The result is

$$v_{\text{av}} = \sqrt{\frac{8kT}{\pi m}} \quad (18.35)$$

Finally, to find the rms speed, we substitute Eq. (18.32) into Eq. (18.31). Evaluating the resulting integral takes some mathematical acrobatics, but we can find it in a table of integrals. The result is

$$v_{\text{rms}} = \sqrt{\frac{3kT}{m}} \quad (18.36)$$

This result agrees with Eq. (18.19); it *must* agree if the Maxwell–Boltzmann distribution is to be consistent with our kinetic-theory calculations.

Table 18.2 shows the fraction of all the molecules in an ideal gas that have speeds *less than* various multiples of v_{rms} . These numbers were obtained by numerical integration; they are the same for all ideal gases.

The distribution of molecular speeds in liquids is similar, although not identical, to that for gases. We can understand evaporation and the vapor pressure of a liquid on this basis. Suppose a molecule must have a speed at least as great as v_A in Fig. 18.23b to escape from the surface of a liquid into the adjacent vapor. The number of such molecules, represented by the area under the “tail” of each curve (to the right of v_A), increases rapidly with temperature. Thus the rate at which molecules can escape is strongly temperature-dependent. This process is balanced by one in which molecules in the vapor phase collide inelastically with the surface and are trapped into the liquid phase. The number of molecules suffering this fate per unit time is proportional to the pressure in the vapor phase. Phase equilibrium between liquid and vapor occurs when these two competing processes proceed at the same rate. So if the molecular speed distributions are known for various temperatures, we can make a theoretical prediction of vapor pressure as a function of temperature. When liquid evaporates, it’s the high-speed molecules that escape from the surface. The ones that are left have less energy on average; this gives us a molecular view of evaporative cooling.

Rates of chemical reactions are often strongly temperature-dependent, and the reason is contained in the Maxwell–Boltzmann distribution. When two reacting molecules collide, the reaction can occur only when the molecules are close enough for their electrons to interact strongly. This requires a minimum energy, called the *activation energy*, and thus a minimum molecular speed. Figure 18.23a shows that the number of molecules in the high-speed tail of the curve increases rapidly with temperature. Thus we expect the rate of any reaction with an activation energy to increase rapidly with temperature.

TEST YOUR UNDERSTANDING OF SECTION 18.5 A quantity of gas containing N molecules has a speed distribution function $f(v)$. How many molecules have speeds between v_1 and $v_2 > v_1$? (i) $\int_0^{v_2} f(v) dv - \int_0^{v_1} f(v) dv$; (ii) $N[\int_0^{v_2} f(v) dv - \int_0^{v_1} f(v) dv]$; (iii) $\int_0^{v_1} f(v) dv - \int_0^{v_2} f(v) dv$; (iv) $N[\int_0^{v_1} f(v) dv - \int_0^{v_2} f(v) dv]$; (v) none of these.

ANSWER (ii) Figure 18.23b shows that the fraction of molecules with speeds between v_1 and v_2 equals the area under the curve of $f(v)$ versus v from v_1 to v_2 . This is equal to the integral $\int_{v_1}^{v_2} f(v) dv$, which in turn is equal to the difference between the integrals $\int_0^{v_2} f(v) dv$ and $\int_0^{v_1} f(v) dv$ (the fraction of molecules with speeds from 0 to v_2 minus the fraction of molecules with speeds from 0 to v_1). The number of molecules with speeds from v_1 to v_2 equals the fraction of molecules in this speed range multiplied by N , the total number of molecules.

TABLE 18.2 Fractions of Molecules in an Ideal Gas with Speeds Less Than Various Multiples of v_{rms}

v/v_{rms}	Fraction
0.20	0.011
0.40	0.077
0.60	0.218
0.80	0.411
1.00	0.608
1.20	0.771
1.40	0.882
1.60	0.947
1.80	0.979
2.00	0.993

BIO APPLICATION Activation Energy and Moth Activity

This hawkmoth of genus *Manduca* cannot fly if the temperature of its muscles is below 29°C. The reason is that the enzyme-catalyzed reactions that power aerobic metabolism and enable muscle action require a minimum molecular energy (activation energy). Just like the molecules in an ideal gas, at low temperatures very few of the molecules involved in these reactions have high energy. As the temperature increases, more molecules have the required minimum energy and the reactions take place at a greater rate. Above 29°C, enough power is generated to allow the hawkmoth to fly.



18.6 PHASES OF MATTER

An ideal gas is the simplest system to analyze from a molecular viewpoint because we ignore the interactions between molecules. But those interactions are the very thing that makes matter condense into the liquid and solid phases under some conditions. So it’s not surprising that theoretical analysis of liquid and solid structure and behavior is a lot more

Figure 18.24 A typical pT phase diagram, showing regions of temperature and pressure at which the various phases exist and where phase changes occur.

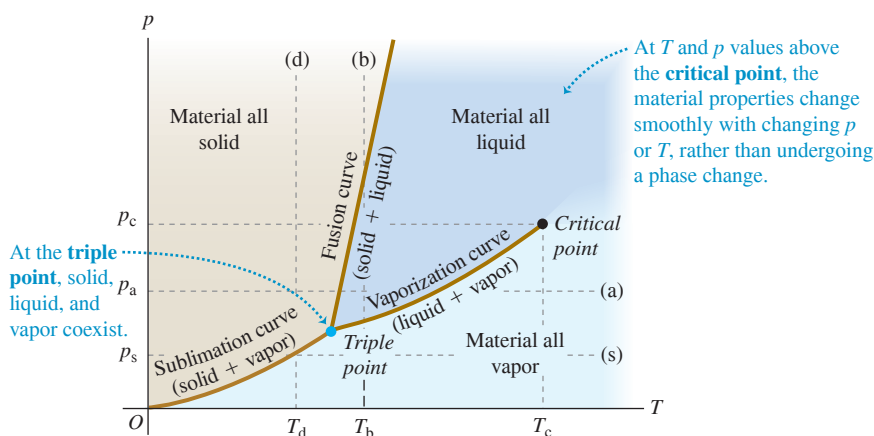


Figure 18.25 Atmospheric pressure on earth is higher than the triple-point pressure of water (see line (a) in Fig. 18.24). Depending on the temperature, water can exist as a vapor (in the atmosphere), as a liquid (in the ocean), or as a solid (like the iceberg shown here).



complicated than that for gases. We won't try to go far here with a microscopic picture, but we can talk in general about phases of matter, phase equilibrium, and phase transitions.

In Section 17.6 we learned that each phase is stable in only certain ranges of temperature and pressure. A transition from one phase to another ordinarily requires **phase equilibrium** between the two phases, and for a given pressure this occurs at only one specific temperature. We can represent these conditions on a graph with axes p and T , called a **phase diagram**; Fig. 18.24 shows an example. Each point on the diagram represents a pair of values of p and T .

Only a single phase can exist at each point in Fig. 18.24, except for points on the solid lines, where two phases can coexist in phase equilibrium. The fusion curve separates the solid and liquid areas and represents possible conditions of solid-liquid phase equilibrium. The vaporization curve separates the liquid and vapor areas, and the sublimation curve separates the solid and vapor areas. All three curves meet at the **triple point**, the only condition under which all three phases can coexist (Fig. 18.25). In Section 17.3 we used the triple-point temperature of water to define the Kelvin temperature scale. Table 18.3 gives triple-point data for several substances.

If we heat a substance at a constant pressure p_a , it goes through a series of states represented by the horizontal line (a) in Fig. 18.24. The melting and boiling temperatures at this pressure are the temperatures at which the line intersects the fusion and vaporization curves, respectively. When the pressure is p_s , constant-pressure heating transforms a substance from solid directly to vapor. This process is called *sublimation*; the intersection of line (s) with the sublimation curve gives the temperature T_s at which it occurs for a pressure p_s . At any pressure less than the triple-point pressure, no liquid phase is possible. The triple-point pressure for carbon dioxide (CO_2) is 5.1 atm. At normal atmospheric pressure, solid CO_2 (“dry ice”) undergoes sublimation; there is no liquid phase.

Line (b) in Fig. 18.24 represents compression at a constant temperature T_b . The material passes from vapor to liquid and then to solid at the points where line (b) crosses the vaporization curve and fusion curve, respectively. Line (d) shows constant-temperature compression at a lower temperature T_d ; the material passes from vapor to solid at the point where line (d) crosses the sublimation curve.

We saw in the pV -diagram of Fig. 18.7 that a liquid-vapor phase transition occurs only when the temperature and pressure are less than those at the point at the top of the green shaded area labeled “Liquid-vapor phase equilibrium region.” This point corresponds to the endpoint at the top of the vaporization curve in Fig. 18.24. It is called the **critical point**, and the corresponding values of p and T are called the critical pressure and temperature, p_c and T_c . A gas at a pressure *above* the critical pressure does not separate into two phases when it is cooled at constant pressure (along a horizontal line above the critical point in Fig. 18.24). Instead, its properties change gradually and continuously from those we ordinarily associate with a gas (low density, large compressibility) to those of a liquid (high density, small compressibility) *without a phase transition*.

TABLE 18.3 Triple-Point Data

Substance	Temperature (K)	Pressure (Pa)
Hydrogen	13.80	0.0704×10^5
Deuterium	18.63	0.171×10^5
Neon	24.56	0.432×10^5
Nitrogen	63.18	0.125×10^5
Oxygen	54.36	0.00152×10^5
Ammonia	195.40	0.0607×10^5
Carbon dioxide	216.55	5.17×10^5
Sulfur dioxide	197.68	0.00167×10^5
Water	273.16	0.00610×10^5

You can understand this by thinking about liquid-phase transitions at successively higher points on the vaporization curve. As we approach the critical point, the *differences* in physical properties (such as density and compressibility) between the liquid and vapor phases become smaller. Exactly *at* the critical point they all become zero, and at this point the distinction between liquid and vapor disappears. The heat of vaporization also grows smaller as we approach the critical point, and it too becomes zero at the critical point.

For nearly all familiar materials the critical pressures are much greater than atmospheric pressure, so we don't observe this behavior in everyday life. For example, the critical point for water is at 647.4 K and 221.2×10^5 Pa (about 218 atm or 3210 psi). But high-pressure steam boilers in electric generating plants regularly run at pressures and temperatures well above the critical point.

Many substances can exist in more than one solid phase. A familiar example is carbon, which exists as noncrystalline soot and crystalline graphite and diamond. Water is another example; more than a dozen types of ice, differing in crystal structure and physical properties, have been observed at very high pressures.

***pVT*-Surfaces**

We remarked in Section 18.1 that for any material, it can be useful to represent the equation of state as a surface in a three-dimensional space with coordinates p , V , and T . **Figure 18.26** shows a typical pVT -surface. The light lines represent pV -isotherms; projecting them onto the pV -plane gives a diagram similar to Fig. 18.7. The pV -isotherms represent contour lines on the pVT -surface, just as contour lines on a topographic map represent the elevation (the third dimension) at each point. The projections of the edges of the surface onto the pT -plane give the pT phase diagram of Fig. 18.24.

Line $abcdef$ in Fig. 18.26 represents constant-pressure heating, with melting along bc and vaporization along de . Note the volume changes that occur as T increases along this line. Line $ghijklm$ corresponds to an isothermal (constant temperature) compression, with liquefaction along hj and solidification along kl . Between these, segments gh and jk represent isothermal compression with increase in pressure; the pressure increases are much greater in the liquid region jk and the solid region lm than in the vapor region gh . Finally, line $nopq$ represents isothermal solidification directly from vapor, as in the formation of snowflakes or frost.

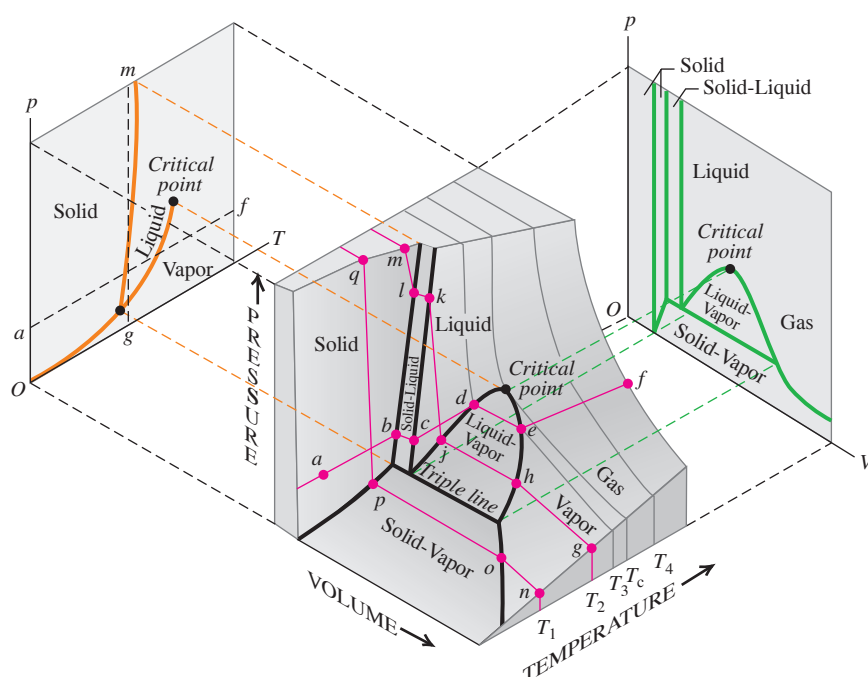


Figure 18.26 A pVT -surface for a substance that expands on melting. Projections of the boundaries on the surface onto the pT - and pV -planes are also shown.

Figure 18.27 A pVT -surface for an ideal gas. At the left, each orange line corresponds to a certain constant volume; at the right, each green line corresponds to a certain constant temperature.

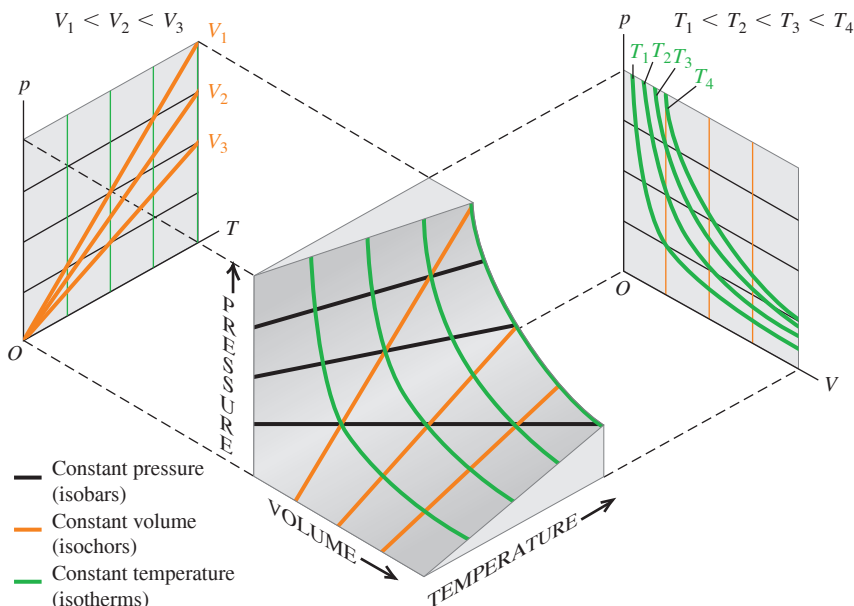


Figure 18.27 shows the much simpler pVT -surface for a substance that obeys the ideal-gas equation of state under all conditions. The projections of the constant-temperature curves onto the pV -plane correspond to the curves of Fig. 18.6, and the projections of the constant-volume curves onto the pT -plane show that pressure is directly proportional to absolute temperature. Figure 18.27 also shows the *isobars* (curves of constant pressure) and *isochors* (curves of constant volume) for an ideal gas.

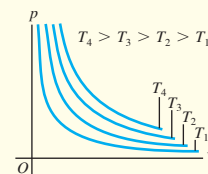
TEST YOUR UNDERSTANDING OF SECTION 18.6 The average atmospheric pressure on Mars is 6.0×10^2 Pa. Could there be lakes of liquid water on the surface of Mars today? What about in the past, when the atmospheric pressure is thought to have been substantially greater?

ANSWER **no, yes** The triple-point pressure of water from Table 18.3 is 6.10×10^2 Pa. The present-day pressure on Mars is just less than this value, corresponding to the line labeled p_s in Fig. 18.24. Hence liquid water cannot exist on the present-day Martian surface, and there are no rivers or lakes. Planetary scientists conclude that liquid water could have existed and almost certainly did exist on Mars in the past, when the atmosphere was thicker.

CHAPTER 18 SUMMARY

Equations of state: The pressure p , volume V , and absolute temperature T of a given quantity of a substance are related by an equation of state. This relationship applies only for equilibrium states, in which p and T are uniform throughout the system. The ideal-gas equation of state, Eq. (18.3), involves the number of moles n and a constant R that is the same for all gases. (See Examples 18.1–18.4.)

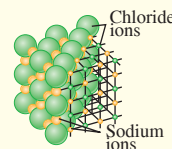
$$pV = nRT \quad (18.3)$$



Molecular properties of matter: The molar mass M of a pure substance is the mass per mole. The mass m_{total} of a quantity of substance equals M multiplied by the number of moles n . Avogadro's number N_A is the number of molecules in a mole. The mass m of an individual molecule is M divided by N_A . (See Example 18.5.)

$$m_{\text{total}} = nM \quad (18.2)$$

$$M = N_A m \quad (18.6)$$



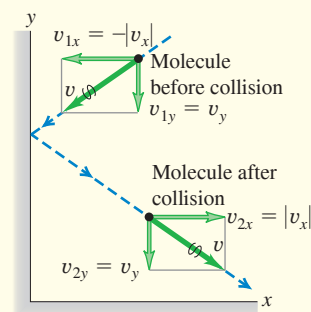
Kinetic-molecular model of an ideal gas: In an ideal gas, the total translational kinetic energy of the gas as a whole (K_{tr}) and the average translational kinetic energy per molecule [$\frac{1}{2}m(v^2)_{av}$] are proportional to the absolute temperature T , and the root-mean-square speed of molecules is proportional to the square root of T . These expressions involve the Boltzmann constant $k = R/N_A$. (See Examples 18.6 and 18.7.) The mean free path λ of molecules in an ideal gas depends on the number of molecules per volume (N/V) and the molecular radius r . (See Example 18.8.)

$$K_{tr} = \frac{3}{2}nRT \quad (18.14)$$

$$\frac{1}{2}m(v^2)_{av} = \frac{3}{2}kT \quad (18.16)$$

$$v_{rms} = \sqrt{(v^2)_{av}} = \sqrt{\frac{3kT}{m}} \quad (18.19)$$

$$\lambda = v t_{mean} = \frac{V}{4\pi\sqrt{2}r^2N} \quad (18.21)$$

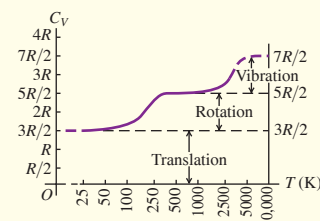


Heat capacities: The molar heat capacity at constant volume C_V is a simple multiple of the gas constant R for certain idealized cases: an ideal monatomic gas [Eq. (18.25)]; an ideal diatomic gas including rotational energy [Eq. (18.26)]; and an ideal monatomic solid [Eq. (18.28)]. Many real systems are approximated well by these idealizations.

$$C_V = \frac{3}{2}R \text{ (monatomic gas)} \quad (18.25)$$

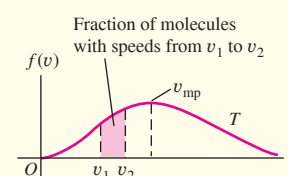
$$C_V = \frac{5}{2}R \text{ (diatomic gas)} \quad (18.26)$$

$$C_V = 3R \text{ (monatomic solid)} \quad (18.28)$$

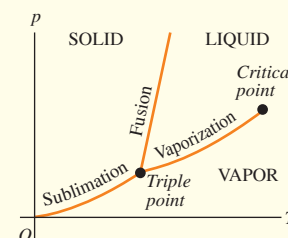


Molecular speeds: The speeds of molecules in an ideal gas are distributed according to the Maxwell-Boltzmann distribution $f(v)$. The quantity $f(v) dv$ describes what fraction of the molecules have speeds between v and $v + dv$.

$$f(v) = 4\pi \left(\frac{m}{2\pi kT} \right)^{3/2} v^2 e^{-mv^2/2kT} \quad (18.32)$$



Phases of matter: Ordinary matter exists in the solid, liquid, and gas phases. A phase diagram shows conditions under which two phases can coexist in phase equilibrium. All three phases can coexist at the triple point. The vaporization curve ends at the critical point, above which the distinction between the liquid and gas phases disappears.



Chapter 18 Media Assets



GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review **EXAMPLES 18.1, 18.2, 18.3, and 18.4** (Section 18.1) before attempting these problems.

VP18.4.1 When the temperature is 30.0°C , the pressure of the air inside a bicycle tire of fixed volume $1.40 \times 10^{-3} \text{ m}^3$ is $5.00 \times 10^5 \text{ Pa}$. (a) What will be the pressure inside the tire when the temperature drops to 10.0°C ? (b) How many moles of air are inside the tire?

VP18.4.2 When a research balloon is released at sea level, where the temperature is 15.0°C and the atmospheric pressure is $1.01 \times 10^5 \text{ Pa}$, the helium in it has volume 13.0 m^3 . (a) When the balloon reaches an altitude of 32.0 km , where the temperature is -44.5°C and the pressure is 868 Pa , what is the volume of the helium? (b) If this balloon is spherical, how many times larger is its radius at 32.0 km than at sea level?

VP18.4.3 The dwarf planet Pluto has a very thin atmosphere made up almost entirely of nitrogen (N_2 , molar mass $2.8 \times 10^{-2} \text{ kg/mol}$). At Pluto's surface the temperature is 42 K and the atmospheric pressure is 1.0 Pa . At the surface, (a) how many moles of gas are there per cubic meter of atmosphere, and (b) what is the density of the atmosphere in kg/m^3 ?

(For comparison, the values are 42 mol/m^3 and 1.2 kg/m^3 at the earth's surface.)

VP18.4.4 When the pressure on n moles of helium gas is suddenly changed from an initial value of p_1 to a final value of p_2 , the density of the gas changes from its initial value of ρ_1 to a final value of $\rho_2 = \rho_1(p_2/p_1)^{3/5}$. (a) If the initial absolute temperature of the gas is T_1 , what is its final absolute temperature T_2 in terms of T_1 , p_1 , and p_2 ? (b) If the final pressure is 0.500 times the initial pressure, what are the ratio of the final density to the initial density and the ratio of the final temperature to the initial temperature? (c) Repeat part (b) if the final pressure is 2.00 times the initial pressure.

Be sure to review **EXAMPLES 18.6 and 18.7** (Section 18.3) before attempting these problems.

VP18.7.1 At what temperature (in $^\circ\text{C}$) is the rms speed of helium atoms (molar mass 4.00 g/mol) the same as the rms speed of nitrogen molecules (molar mass 28.0 g/mol) at 20.0°C ? (Note that helium remains a gas at temperatures above -269°C .)

Continued

VP18.7.2 For the first 380,000 years after the Big Bang, the temperature of the matter in the universe was too high for nuclei and electrons to form atoms. The first hydrogen atoms (mass per atom 1.67×10^{-27} kg) did not form until the temperature had dropped to about 3000 K. (a) What was the average translational kinetic energy of hydrogen atoms when the temperature was 3.0×10^3 K? (b) What was the rms speed of these atoms?

VP18.7.3 The air in a room with dimensions $5.0 \text{ m} \times 5.0 \text{ m} \times 2.4 \text{ m}$ is at temperature 20.0°C and pressure $1.00 \text{ atm} = 1.01 \times 10^5 \text{ Pa}$. (a) How many air molecules are in this room? (b) What is the total translational kinetic energy of these molecules? (c) How fast would a car of mass 1.5×10^3 kg have to move to have the same translational kinetic energy?

VP18.7.4 (a) What is the average value of the first 10 integers (1 through 10)? (b) What is the average value of the squares of the first 10 integers? (c) What is the rms value of the first 10 integers?

Be sure to review EXAMPLE 18.8 (Section 18.3) before attempting these problems.

VP18.8.1 At the surface of Mars the atmosphere has average pressure $6.0 \times 10^2 \text{ Pa}$ and average temperature -63°C . (a) What is the mean free path of atmospheric molecules (assumed to be spheres of radius $2.0 \times 10^{-10} \text{ m}$) at the surface of Mars? (b) How many times greater is

your answer in part (a) than the mean free path of atmospheric molecules at the earth's surface, where the average pressure is $1.01 \times 10^5 \text{ Pa}$ and the average temperature is 15°C ?

VP18.8.2 A cubical box 1.00 m on a side contains air at 20.0°C . (a) What would the pressure inside the box have to be in order for the mean free path of air molecules (assumed to be spheres of radius $2.0 \times 10^{-10} \text{ m}$) to be 1.00 m , so that a typical molecule suffers no collisions with other molecules as it travels the width of the box? (b) How many moles of air would be inside the box at the pressure calculated in part (a)? (At 1 atm pressure, this box would contain 41.4 mol of air.)

VP18.8.3 A cylinder for storing helium (molar mass 4.00 g/mol) has an interior volume of $50.0 \text{ L} = 5.00 \times 10^{-2} \text{ m}^3$. The cylinder holds $4.00 \times 10^2 \text{ mol}$ of helium under pressure at 27.0°C . If you assume the helium atoms are spheres of radius $3.1 \times 10^{-11} \text{ m}$, what are (a) the mean free path of a helium atom in the tank and (b) the mean free time for a helium atom moving at the rms speed?

VP18.8.4 A gas with molecules of radius r and mass per molecule m is at temperature T and pressure p . (a) Write an expression for the mean free time for a molecule moving at the rms speed for this gas. (b) Which single change would have the greatest effect on the mean free time: doubling the radius r , doubling the pressure p , or doubling the temperature T ?

BRIDGING PROBLEM Gas on Jupiter's Moon Europa

An astronaut visiting Jupiter's satellite Europa leaves a canister of 1.20 mol of nitrogen gas (28.0 g/mol) at 25.0°C on the satellite's surface. Europa has no significant atmosphere, and the acceleration due to gravity at its surface is 1.30 m/s^2 . The canister springs a leak, allowing molecules to escape from a small hole. (a) What is the maximum height (in km) above Europa's surface that is reached by a nitrogen molecule whose speed equals the rms speed? Assume that the molecule is shot straight up out of the hole in the canister, and ignore the variation in g with altitude. (b) The escape speed from Europa is 2025 m/s . Can any of the nitrogen molecules escape from Europa and into space?

SOLUTION GUIDE

IDENTIFY and SET UP

1. Sketch the situation, showing all relevant dimensions.
2. List the unknown quantities, and decide which are the target variables.
3. How will you find the rms speed of the nitrogen molecules? What principle will you use to find the maximum height that a molecule with this speed can reach?

4. Does the rms molecular speed in the gas represent the maximum molecular speed? If not, what is the maximum speed?

EXECUTE

5. Solve for the rms speed. Use this to calculate the maximum height that a molecule with this speed can reach.
6. Use your result from step 5 to answer the question in part (b).

EVALUATE

7. Do your results depend on the amount of gas in the container? Why or why not?
8. How would your results from steps 5 and 6 be affected if the gas cylinder were instead left on Jupiter's satellite Ganymede, which has higher surface gravity than Europa and a higher escape speed? Like Europa, Ganymede has no significant atmosphere.

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q18.1 Section 18.1 states that ordinarily, pressure, volume, and temperature cannot change individually without one affecting the others. Yet when a liquid evaporates, its volume changes, even though its pressure and temperature are constant. Is this inconsistent? Why or why not?

Q18.2 In the ideal-gas equation, could an equivalent Celsius temperature be used instead of the Kelvin one if an appropriate numerical value of the constant R is used? Why or why not?

Q18.3 When a car is driven some distance, the air pressure in the tires increases. Why? Should you let out some air to reduce the pressure? Why or why not?

Q18.4 The coolant in an automobile radiator is kept at a pressure higher than atmospheric pressure. Why is this desirable? The radiator cap will release coolant when the gauge pressure of the coolant reaches a certain value, typically 15 lb/in.^2 or so. Why not just seal the system completely?

Q18.5 Unwrapped food placed in a freezer experiences dehydration, known as "freezer burn." Why?

Q18.6 A group of students drove from their university (near sea level) up into the mountains for a skiing weekend. Upon arriving at the slopes, they discovered that the bags of potato chips they had brought for snacks had all burst open. What caused this to happen?

Q18.7 The derivation of the ideal-gas equation included the assumption that the number of molecules is very large, so that we could compute the average force due to many collisions. However, the ideal-gas equation holds accurately only at low pressures, where the molecules are few and far between. Is this inconsistent? Why or why not?

Q18.8 A rigid, perfectly insulated container has a membrane dividing its volume in half. One side contains a gas at an absolute temperature T_0 and pressure p_0 , while the other half is completely empty. Suddenly a small hole develops in the membrane, allowing the gas to leak out into the other half until it eventually occupies twice its original volume. In terms of T_0 and p_0 , what will be the new temperature and pressure of the gas when it is distributed equally in both halves of the container? Explain your reasoning.

Q18.9 (a) Which has more atoms: a kilogram of hydrogen or a kilogram of lead? Which has more mass? (b) Which has more atoms: a mole of hydrogen or a mole of lead? Which has more mass? Explain your reasoning.

Q18.10 Use the concepts of the kinetic-molecular model to explain: (a) why the pressure of a gas in a rigid container increases as heat is added to the gas and (b) why the pressure of a gas increases as we compress it, even if we do not change its temperature.

Q18.11 The proportions of various gases in the earth's atmosphere change somewhat with altitude. Would you expect the proportion of oxygen at high altitude to be greater or less than at sea level compared to the proportion of nitrogen? Why?

Q18.12 Comment on the following statement: *When two gases are mixed, if they are to be in thermal equilibrium, they must have the same average molecular speed.* Is the statement correct? Why or why not?

Q18.13 The kinetic-molecular model contains a hidden assumption about the temperature of the container walls. What is this assumption? What would happen if this assumption were not valid?

Q18.14 The temperature of an ideal gas is directly proportional to the average kinetic energy of its molecules. If a container of ideal gas is moving past you at 2000 m/s, is the temperature of the gas higher than if the container was at rest? Explain your reasoning.

Q18.15 If the pressure of an ideal monatomic gas is increased while the number of moles is kept constant, what happens to the average translational kinetic energy of one atom of the gas? Is it possible to change *both* the volume and the pressure of an ideal gas and keep the average translational kinetic energy of the atoms constant? Explain.

Q18.16 In deriving the ideal-gas equation from the kinetic-molecular model, we ignored potential energy due to the earth's gravity. Is this omission justified? Why or why not?

Q18.17 Imagine a special air filter placed in a window of a house. The tiny holes in the filter allow only air molecules moving faster than a certain speed to exit the house, and allow only air molecules moving slower than that speed to enter the house from outside. What effect would this filter have on the temperature inside the house? (It turns out that the second law of thermodynamics—which we'll discuss in Chapter 20—tells us that such a wonderful air filter would be impossible to make.)

Q18.18 A gas storage tank has a small leak. The pressure in the tank drops more quickly if the gas is hydrogen or helium than if it is oxygen. Why?

Q18.19 Consider two specimens of ideal gas at the same temperature. Specimen A has the same total mass as specimen B, but the molecules in specimen A have greater molar mass than they do in specimen B. In which specimen is the total kinetic energy of the gas greater? Does your answer depend on the molecular structure of the gases? Why or why not?

Q18.20 The temperature of an ideal monatomic gas is increased from 25°C to 50°C. Does the average translational kinetic energy of each gas atom double? Explain. If your answer is no, what would the final temperature be if the average translational kinetic energy was doubled?

Q18.21 If the root-mean-square speed of the atoms of an ideal gas is to be doubled, by what factor must the Kelvin temperature of the gas be increased? Explain.

Q18.22 (a) If you apply the same amount of heat to 1.00 mol of an ideal monatomic gas and 1.00 mol of an ideal diatomic gas, which one (if any) will increase more in temperature? (b) Physically, *why* do diatomic gases have a greater molar heat capacity than monatomic gases?

Q18.23 The discussion in Section 18.4 concluded that all ideal monatomic gases have the same heat capacity C_V . Does this mean that it takes the same amount of heat to raise the temperature of 1.0 g of each one by 1.0 K? Explain your reasoning.

Q18.24 In a gas that contains N molecules, is it accurate to say that the number of molecules with speed v is equal to $f(v)$? Is it accurate to say that this number is given by $Nf(v)$? Explain your answers.

Q18.25 The atmosphere of the planet Mars is 95.3% carbon dioxide (CO_2) and about 0.03% water vapor. The atmospheric pressure is only about 600 Pa, and the surface temperature varies from -30°C to -100°C . The polar ice caps contain both CO_2 ice and water ice. Could there be *liquid* CO_2 on the surface of Mars? Could there be liquid water? Why or why not?

Q18.26 A beaker of water at room temperature is placed in an enclosure, and the air pressure in the enclosure is slowly reduced. When the air pressure is reduced sufficiently, the water begins to boil. The temperature of the water does not rise when it boils; in fact, the temperature *drops* slightly. Explain these phenomena.

Q18.27 When you stand on ice with skates, the blades on the bottom have a low surface area, and so they pressurize the ice beneath the blades. When ice is pressurized, is the melting temperature greater than or less than 0°C ? Explain.

Q18.28 Hydrothermal vents are openings in the ocean floor that discharge very hot water. The water emerging from one such vent off the Oregon coast, 2400 m below the surface, is at 279°C . Despite its high temperature, the water doesn't boil. Why not?

Q18.29 The dark areas on the moon's surface are called *maria*, Latin for "seas," and were once thought to be bodies of water. In fact, the maria are not "seas" at all, but plains of solidified lava. Given that there is no atmosphere on the moon, how can you explain the absence of liquid water on the moon's surface?

Q18.30 In addition to the normal cooking directions printed on the back of a box of rice, there are also "high-altitude directions." The only difference is that the "high-altitude directions" suggest increasing the cooking time and using a greater volume of boiling water in which to cook the rice. Why should the directions depend on the altitude in this way?

EXERCISES

Section 18.1 Equations of State

18.1 • A 20.0 L tank contains 4.86×10^{-4} kg of helium at 18.0°C . The molar mass of helium is 4.00 g/mol. (a) How many moles of helium are in the tank? (b) What is the pressure in the tank, in pascals and in atmospheres?

18.2 •• Helium gas with a volume of 3.20 L, under a pressure of 0.180 atm and at 41.0°C , is warmed until both pressure and volume are doubled. (a) What is the final temperature? (b) How many grams of helium are there? The molar mass of helium is 4.00 g/mol.

18.3 • A cylindrical tank has a tight-fitting piston that allows the volume of the tank to be changed. The tank originally contains 0.110 m^3 of air at a pressure of 0.355 atm. The piston is slowly pulled out until the volume of the gas is increased to 0.390 m^3 . If the temperature remains constant, what is the final value of the pressure?

18.4 • A 3.00 L tank contains air at 3.00 atm and 20.0°C . The tank is sealed and cooled until the pressure is 1.00 atm. (a) What is the temperature then in degrees Celsius? Assume that the volume of the tank is constant. (b) If the temperature is kept at the value found in part (a) and the gas is compressed, what is the volume when the pressure again becomes 3.00 atm?

18.5 • The discussion following Eq. (18.7) gives the constants in the van der Waals equation for CO_2 gas. It also says that at STP the van der Waals equation gives only a small (0.5%) correction in the ideal-gas equation. Consider 1 mole of CO_2 gas at $T = 273.0 \text{ K}$ and a volume of $4.48 \times 10^{-4} \text{ m}^3$. (a) What is the pressure of the gas calculated by the ideal-gas equation? (b) What does the van der Waals equation give for the pressure? What is the percentage difference from the ideal-gas result?

18.6 •• You have several identical balloons. You experimentally determine that a balloon will break if its volume exceeds 0.900 L . The pressure of the gas inside the balloon equals air pressure (1.00 atm). (a) If the air inside the balloon is at a constant 22.0°C and behaves as an ideal gas, what mass of air can you blow into one of the balloons before it bursts? (b) Repeat part (a) if the gas is helium rather than air.

18.7 •• A Jaguar XK8 convertible has an eight-cylinder engine. At the beginning of its compression stroke, one of the cylinders contains 499 cm^3 of air at atmospheric pressure ($1.01 \times 10^5 \text{ Pa}$) and a temperature of 27.0°C . At the end of the stroke, the air has been compressed to a volume of 46.2 cm^3 and the gauge pressure has increased to $2.72 \times 10^6 \text{ Pa}$. Compute the final temperature.

18.8 •• A welder using a tank of volume 0.0750 m^3 fills it with oxygen (molar mass 32.0 g/mol) at a gauge pressure of $3.00 \times 10^5 \text{ Pa}$ and temperature of 37.0°C . The tank has a small leak, and in time some of the oxygen leaks out. On a day when the temperature is 22.0°C , the gauge pressure of the oxygen in the tank is $1.80 \times 10^5 \text{ Pa}$. Find (a) the initial mass of oxygen and (b) the mass of oxygen that has leaked out.

18.9 •• A large cylindrical tank contains 0.750 m^3 of nitrogen gas at 27°C and $7.50 \times 10^3 \text{ Pa}$ (absolute pressure). The tank has a tight-fitting piston that allows the volume to be changed. What will be the pressure if the volume is decreased to 0.410 m^3 and the temperature is increased to 157°C ?

18.10 • An empty cylindrical canister 1.50 m long and 90.0 cm in diameter is to be filled with pure oxygen at 22.0°C to store in a space station. To hold as much gas as possible, the absolute pressure of the oxygen will be 21.0 atm . The molar mass of oxygen is 32.0 g/mol . (a) How many moles of oxygen does this canister hold? (b) For someone lifting this canister, by how many kilograms does this gas increase the mass to be lifted?

18.11 • The gas inside a balloon will always have a pressure nearly equal to atmospheric pressure, since that is the pressure applied to the outside of the balloon. You fill a balloon with helium (a nearly ideal gas) to a volume of 0.600 L at 19.0°C . What is the volume of the balloon if you cool it to the boiling point of liquid nitrogen (77.3 K)?

18.12 • An ideal gas has a density of $1.33 \times 10^{-6} \text{ g/cm}^3$ at $1.00 \times 10^{-3} \text{ atm}$ and 20.0°C . Identify the gas.

18.13 •• If a certain amount of ideal gas occupies a volume V at STP on earth, what would be its volume (in terms of V) on Venus, where the temperature is 1003°C and the pressure is 92 atm ?

18.14 • A diver observes a bubble of air rising from the bottom of a lake (where the absolute pressure is 3.50 atm) to the surface (where the pressure is 1.00 atm). The temperature at the bottom is 4.0°C , and the temperature at the surface is 23.0°C . (a) What is the ratio of the volume of the bubble as it reaches the surface to its volume at the bottom? (b) Would it be safe for the diver to hold his breath while ascending from the bottom of the lake to the surface? Why or why not?

18.15 • How many moles of an ideal gas exert a gauge pressure of 0.876 atm in a volume of 5.43 L at a temperature of 22.2°C ?

18.16 • Three moles of an ideal gas are in a rigid cubical box with sides of length 0.300 m . (a) What is the force that the gas exerts on each of the six sides of the box when the gas temperature is 20.0°C ? (b) What is the force when the temperature of the gas is increased to 100.0°C ?

18.17 •• (a) Calculate the mass of nitrogen present in a volume of 3000 cm^3 if the gas is at 22.0°C and the absolute pressure of $2.00 \times 10^{-13} \text{ atm}$ is a partial vacuum easily obtained in laboratories. (b) What is the density (in kg/m^3) of the N_2 ?

18.18 • At an altitude of $11,000 \text{ m}$ (a typical cruising altitude for a jet airliner), the air temperature is -56.5°C and the air density is 0.364 kg/m^3 . What is the pressure of the atmosphere at that altitude? (Note: The temperature at this altitude is not the same as at the surface of the earth, so the calculation of Example 18.4 in Section 18.1 doesn't apply.)

Section 18.2 Molecular Properties of Matter

18.19 • How many moles are in a 1.00 kg bottle of water? How many molecules? The molar mass of water is 18.0 g/mol .

18.20 • A large organic molecule has a mass of $1.41 \times 10^{-21} \text{ kg}$. What is the molar mass of this compound?

18.21 •• Modern vacuum pumps make it easy to attain pressures of the order of 10^{-13} atm in the laboratory. Consider a volume of air and treat the air as an ideal gas. (a) At a pressure of $9.00 \times 10^{-14} \text{ atm}$ and an ordinary temperature of 300.0 K , how many molecules are present in a volume of 1.00 cm^3 ? (b) How many molecules would be present at the same temperature but at 1.00 atm instead?

18.22 •• The Lagoon Nebula Figure E18.22

(Fig. E18.22) is a cloud of hydrogen gas located 3900 light-years from the earth. The cloud is about 45 light-years in diameter and glows because of its high temperature of 7500 K . (The gas is raised to this temperature by the stars that lie within the nebula.) The cloud is also very thin; there are only 80



molecules per cubic centimeter. (a) Find the gas pressure (in atmospheres) in the Lagoon Nebula. Compare it to the laboratory pressure referred to in Exercise 18.21. (b) Science-fiction films sometimes show starships being buffeted by turbulence as they fly through gas clouds such as the Lagoon Nebula. Does this seem realistic? Why or why not?

18.23 •• **How Close Together Are Gas Molecules?** Consider an ideal gas at 27°C and 1.00 atm . To get some idea how close these molecules are to each other, on the average, imagine them to be uniformly spaced, with each molecule at the center of a small cube. (a) What is the length of an edge of each cube if adjacent cubes touch but do not overlap? (b) How does this distance compare with the diameter of a typical molecule? (c) How does their separation compare with the spacing of atoms in solids, which typically are about 0.3 nm apart?

Section 18.3 Kinetic-Molecular Model of an Ideal Gas

18.24 •• A container with rigid walls holds n moles of a monatomic ideal gas. In terms of n , how many moles of the gas must be removed from the container to double the pressure while also doubling the rms speed of the gas atoms?

18.25 • (a) What is the total translational kinetic energy of the air in an empty room that has dimensions $8.00 \text{ m} \times 12.00 \text{ m} \times 4.00 \text{ m}$ if the air is treated as an ideal gas at 1.00 atm ? (b) What is the speed of a 2000 kg automobile if its kinetic energy equals the translational kinetic energy calculated in part (a)?

18.26 • A flask contains a mixture of neon (Ne), krypton (Kr), and radon (Rn) gases. Compare (a) the average kinetic energies of the three types of atoms and (b) the root-mean-square speeds. (Hint: Appendix D shows the molar mass (in g/mol) of each element under the chemical symbol for that element.)

18.27 • A container with volume 80.0 cm^3 and rigid walls holds a monatomic ideal gas. To determine the number of gas atoms in the container, you measure the pressure p_{atm} of the gas in atmospheres as a function of the Celsius temperature T_c of the gas. You plot p_{atm} versus T_c and find that your data lie close to a straight line that has slope 1.10 atm/C° . What is your experimental result for the number of gas atoms?

18.28 • A container with volume 1.64 L is initially evacuated. Then it is filled with 0.226 g of N_2 . Assume that the pressure of the gas is low enough for the gas to obey the ideal-gas law to a high degree of accuracy. If the root-mean-square speed of the gas molecules is 182 m/s, what is the pressure of the gas?

18.29 •• (a) A deuteron, ${}^2_1\text{H}$, is the nucleus of a hydrogen isotope and consists of one proton and one neutron. The plasma of deuterons in a nuclear fusion reactor must be heated to about 300 million K. What is the rms speed of the deuterons? Is this a significant fraction of the speed of light in vacuum ($c = 3.0 \times 10^8$ m/s)? (b) What would the temperature of the plasma be if the deuterons had an rms speed equal to $0.10c$?

18.30 • Martian Climate. The atmosphere of Mars is mostly CO_2 (molar mass 44.0 g/mol) under a pressure of 650 Pa, which we shall assume remains constant. In many places the temperature varies from 0.0°C in summer to -100°C in winter. Over the course of a Martian year, what are the ranges of (a) the rms speeds of the CO_2 molecules and (b) the density (in mol/m^3) of the atmosphere?

18.31 •• Oxygen (O_2) has a molar mass of 32.0 g/mol. What is (a) the average translational kinetic energy of an oxygen molecule at a temperature of 300 K; (b) the average value of the square of its speed; (c) the root-mean-square speed; (d) the momentum of an oxygen molecule traveling at this speed? (e) Suppose an oxygen molecule traveling at this speed bounces back and forth between opposite sides of a cubical vessel 0.10 m on a side. What is the average force the molecule exerts on one of the walls of the container? (Assume that the molecule's velocity is perpendicular to the two sides that it strikes.) (f) What is the average force per unit area? (g) How many oxygen molecules traveling at this speed are necessary to produce an average pressure of 1 atm? (h) Compute the number of oxygen molecules that are contained in a vessel of this size at 300 K and atmospheric pressure. (i) Your answer for part (h) should be three times as large as the answer for part (g). Where does this discrepancy arise?

18.32 •• Calculate the mean free path of air molecules at 3.50×10^{-13} atm and 300 K. (This pressure is readily attainable in the laboratory; see Exercise 18.21.) As in Example 18.8, model the air molecules as spheres of radius 2.0×10^{-10} m.

18.33 •• At what temperature is the root-mean-square speed of nitrogen molecules equal to the root-mean-square speed of hydrogen molecules at 20.0°C ? (*Hint:* Appendix D shows the molar mass (in g/mol) of each element under the chemical symbol for that element. The molar mass of H_2 is twice the molar mass of hydrogen atoms, and similarly for N_2 .)

18.34 • Smoke particles in the air typically have masses on the order of 10^{-16} kg. The Brownian motion (rapid, irregular movement) of these particles, resulting from collisions with air molecules, can be observed with a microscope. (a) Find the root-mean-square speed of Brownian motion for a particle with a mass of 3.00×10^{-16} kg in air at 300 K. (b) Would the root-mean-square speed be different if the particle were in hydrogen gas at the same temperature? Explain.

Section 18.4 Heat Capacities

18.35 •• Three moles of helium gas (molar mass $M = 4.00$ g/mol) are in a rigid container that keeps the volume of the gas constant. Initially the rms speed of the gas atoms is 900 m/s. What is the rms speed of the gas atoms after 2400 J of heat energy is added to the gas?

18.36 •• A rigid container holds 4.00 mol of a monatomic ideal gas that has temperature 300 K. The initial pressure of the gas is 6.00×10^4 Pa. What is the pressure after 6000 J of heat energy is added to the gas?

18.37 • How much heat does it take to increase the temperature of 1.80 mol of an ideal gas by 50.0 K near room temperature if the gas is held at constant volume and is (a) diatomic; (b) monatomic?

18.38 •• Perfectly rigid containers each hold n moles of ideal gas, one being hydrogen (H_2) and the other being neon (Ne). If it takes 300 J of heat to increase the temperature of the hydrogen by 2.50°C , by how many degrees will the same amount of heat raise the temperature of the neon?

18.39 •• (a) Compute the specific heat at constant volume of nitrogen (N_2) gas, and compare it with the specific heat of liquid water. The molar mass of N_2 is 28.0 g/mol. (b) You warm 1.00 kg of water at a constant volume of 1.00 L from 20.0°C to 30.0°C in a kettle. For the same amount of heat, how many kilograms of 20.0°C air would you be able to warm to 30.0°C ? What volume (in liters) would this air occupy at 20.0°C and a pressure of 1.00 atm? Make the simplifying assumption that air is 100% N_2 .

18.40 •• (a) Calculate the specific heat at constant volume of water vapor, assuming the nonlinear triatomic molecule has three translational and three rotational degrees of freedom and that vibrational motion does not contribute. The molar mass of water is 18.0 g/mol. (b) The actual specific heat of water vapor at low pressures is about $2000 \text{ J/kg} \cdot \text{K}$. Compare this with your calculation and comment on the actual role of vibrational motion.

Section 18.5 Molecular Speeds

18.41 • For diatomic carbon dioxide gas (CO_2 , molar mass 44.0 g/mol) at $T = 300$ K, calculate (a) the most probable speed v_{mp} ; (b) the average speed v_{av} ; (c) the root-mean-square speed v_{rms} .

18.42 • For a gas of nitrogen molecules (N_2), what must the temperature be if 94.7% of all the molecules have speeds less than (a) 1500 m/s; (b) 1000 m/s; (c) 500 m/s? Use Table 18.2. The molar mass of N_2 is 28.0 g/mol.

18.43 • The speed of sound for an ideal gas is given by $v = \sqrt{\gamma RT/M}$ [Eq. (16.10)]. We'll see in Chapter 19 that for a monatomic ideal gas, $\gamma = 1.67$. (a) What is the ratio v_{rms}/v ? (b) The average speed v_{av} of the gas atoms is given by Eq. (18.35). What is the ratio v_{av}/v ?

18.44 • In 0.0345 mol of a monatomic ideal gas, how many of the atoms have speeds that are within 20% of the rms speed? (Use Table 18.2.)

Section 18.6 Phases of Matter

18.45 • Solid water (ice) is slowly warmed from a very low temperature. (a) What minimum external pressure p_1 must be applied to the solid if a melting phase transition is to be observed? Describe the sequence of phase transitions that occur if the applied pressure p is such that $p < p_1$. (b) Above a certain maximum pressure p_2 , no boiling transition is observed. What is this pressure? Describe the sequence of phase transitions that occur if $p_1 < p < p_2$.

18.46 • Meteorology. The *vapor pressure* is the pressure of the vapor phase of a substance when it is in equilibrium with the solid or liquid phase of the substance. The *relative humidity* is the partial pressure of water vapor in the air divided by the vapor pressure of water at that same temperature, expressed as a percentage. The air is saturated when the humidity is 100%. (a) The vapor pressure of water at 20.0°C is 2.34×10^3 Pa. If the air temperature is 20.0°C and the relative humidity is 60%, what is the partial pressure of water vapor in the atmosphere (that is, the pressure due to water vapor alone)? (b) Under the conditions of part (a), what is the mass of water in 1.00 m^3 of air? (The molar mass of water is 18.0 g/mol. Assume that water vapor can be treated as an ideal gas.)

PROBLEMS

18.47 •• CP Imagine the sound made when a latex balloon with a diameter of 30 cm pops 2 m from your ear. Estimate the sound intensity level in decibels, using Table 16.2 as a guide. (a) Assuming the duration of the popping event was 100 ms, use the intensity of the popping sound to determine the average power of the pop and the energy released in the pop. This provides an estimate of the energy stored in the balloon prior to the pop. (b) Let T be the ratio of the energy stored in the stretched balloon to its surface area. Use your estimate of the stored energy and the size of the balloon to estimate the value of T . (c) The gauge pressure of a latex balloon depends on T and radius R according to $p_{\text{gauge}} = T/R$. Use this information to estimate the gauge pressure in the balloon.

18.48 • A physics lecture room at 1.00 atm and 27.0°C has a volume of 216 m³. (a) Use the ideal-gas law to estimate the number of air molecules in the room. Assume that all of the air is N₂. Calculate (b) the particle density—that is, the number of N₂ molecules per cubic centimeter—and (c) the mass of the air in the room.

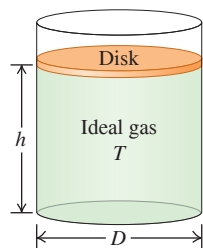
18.49 •• CP BIO The Effect of Altitude on the Lungs. (a) Calculate the change in air pressure you'll experience if you climb a 1000 m mountain, assuming for simplicity that the temperature and air density do not change over this distance and that they were 22°C and 1.2 kg/m³, respectively, at the bottom of the mountain. (Note: The result of Example 18.4 doesn't apply, since the expression derived in that example accounts for the variation of air density with altitude and we are told to ignore that here.) (b) If you took a 0.50 L breath at the foot of the mountain and managed to hold it until you reached the top, what would be the volume of this breath when you exhaled it there?

18.50 •• CP BIO The Bends. If deep-sea divers rise to the surface too quickly, nitrogen bubbles in their blood can expand and prove fatal. This phenomenon is known as the *bends*. If a scuba diver rises quickly from a depth of 25 m in Lake Michigan (which is fresh water), what will be the volume at the surface of an N₂ bubble that occupied 1.0 mm³ in his blood at the lower depth? Does it seem that this difference is large enough to be a problem? (Assume that the pressure difference is due to only the changing water pressure, not to any temperature difference. This assumption is reasonable, since we are warm-blooded creatures.)

18.51 ••• CP A hot-air balloon stays aloft because hot air at atmospheric pressure is less dense than cooler air at the same pressure. If the volume of the balloon is 500.0 m³ and the surrounding air is at 15.0°C, what must the temperature of the air in the balloon be for it to lift a total load of 290 kg (in addition to the mass of the hot air)? The density of air at 15.0°C and atmospheric pressure is 1.23 kg/m³.

18.52 •• In an evacuated enclosure, a vertical cylindrical tank of diameter D is sealed by a 3.00 kg circular disk that can move up and down without friction. Beneath the disk is a quantity of ideal gas at temperature T in the cylinder (Fig. P18.52). Initially the disk is at rest at a distance of $h = 4.00$ m above the bottom of the tank. When a lead brick of mass 9.00 kg is gently placed on the disk, the disk moves downward. If the temperature of the gas is kept constant and no gas escapes from the tank, what distance above the bottom of the tank is the disk when it again comes to rest?

Figure P18.52



18.53 ••• A cylinder 1.00 m tall with inside diameter 0.120 m is used to hold propane gas (molar mass 44.1 g/mol) for use in a barbecue. It is initially filled with gas until the gauge pressure is 1.30×10^6 Pa at 22.0°C. The temperature of the gas remains constant as it is partially emptied out of the tank, until the gauge pressure is 3.40×10^5 Pa. Calculate the mass of propane that has been used.

18.54 • CP During a test dive in 1939, prior to being accepted by the U.S. Navy, the submarine *Squalus* sank at a point where the depth of water was 73.0 m. The temperature was 27.0°C at the surface and 7.0°C at the bottom. The density of seawater is 1030 kg/m³. (a) A diving bell was used to rescue 33 trapped crewmen from the *Squalus*. The diving bell was in the form of a circular cylinder 2.30 m high, open at the bottom and closed at the top. When the diving bell was lowered to the bottom of the sea, to what height did water rise within the diving bell? (Hint: Ignore the relatively small variation in water pressure between the bottom of the bell and the surface of the water within the bell.) (b) At what gauge pressure must compressed air have been supplied to the bell while on the bottom to expel all the water from it?

18.55 •• A parcel of air over a campfire feels an upward buoyant force because the heated air is less dense than the surrounding air. By estimating the acceleration of the air immediately above a fire, one can estimate the fire's temperature. The mass of a volume V of air is nM_{air} , where n is the number of moles of air molecules in the volume and M_{air} is the molar mass of air. The net upward force on a parcel of air above a fire is roughly given by $(m_{\text{out}} - m_{\text{in}})g$, where m_{out} is the mass of a volume of ambient air and m_{in} is the mass of a similar volume of air in the hot zone. (a) Use the ideal-gas law, along with the knowledge that the pressure of the air above the fire is the same as that of the ambient air, to derive an expression for the acceleration a of an air parcel as a function of $(T_{\text{out}}/T_{\text{in}})$, where T_{in} is the absolute temperature of the air above the fire and T_{out} is the absolute temperature of the ambient air. (b) Rearrange your formula from part (a) to obtain an expression for T_{in} as a function of T_{out} and a . (c) Based on your experience with campfires, estimate the acceleration of the air above the fire by comparing in your mind the upward trajectory of sparks with the acceleration of falling objects. Thus you can estimate a as a multiple of g . (d) Assuming an ambient temperature of 15°C, use your formula and your estimate of a to estimate the temperature of the fire.

18.56 • Pressure on Venus. At the surface of Venus the average temperature is a balmy 460°C due to the greenhouse effect (global warming!), the pressure is 92 earth-atmospheres, and the acceleration due to gravity is $0.894g_{\text{earth}}$. The atmosphere is nearly all CO₂ (molar mass 44.0 g/mol), and the temperature remains remarkably constant. Assume that the temperature does not change with altitude. (a) What is the atmospheric pressure 1.00 km above the surface of Venus? Express your answer in Venus-atmospheres and earth-atmospheres. (b) What is the root-mean-square speed of the CO₂ molecules at the surface of Venus and at an altitude of 1.00 km?

18.57 ••• CP CALC A cylindrical diving bell has a radius of 750 cm and a height of 2.50 m. The bell includes a top compartment that holds an undersea adventurer. A bottom compartment separated from the top by a sturdy grating holds a tank of compressed air with a valve to release air into the bell, a second valve that can release air from the bell into the sea, a third valve that regulates the entry of seawater for ballast, a pump that removes the ballast to increase buoyancy, and an electric heater that maintains a constant temperature of 20.0°C. The total mass of the bell and all of its apparatuses is 4350 kg. The density of seawater is 1025 kg/m³. (a) An 80.0 kg adventurer enters the bell. How many liters of seawater should be moved into the bell so that it is neutrally buoyant? (b) By carefully regulating ballast, the bell is made to descend into the sea at a rate of 1.0 m/s. Compressed air is released from the tank to raise the pressure in the bell to match the pressure of the seawater outside the bell. As the bell descends, at what rate should air be released through the first valve? (Hint: Derive an expression for the number of moles of air in the bell n as a function of depth y ; then differentiate this to obtain dn/dt as a function of dy/dt .) (c) If the compressed air tank is a fully loaded, specially designed, 600 ft³ tank, which means it contains that volume of air at standard temperature and pressure (0°C and 1 atm), how deep can the bell descend?

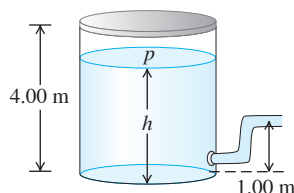
18.58 •• A flask with a volume of 1.50 L, provided with a stopcock, contains ethane gas (C₂H₆) at 300 K and atmospheric pressure (1.013×10^5 Pa). The molar mass of ethane is 30.1 g/mol. The system is warmed to a temperature of 550 K, with the stopcock open to the atmosphere. The stopcock is then closed, and the flask is cooled to its original temperature. (a) What is the final pressure of the ethane in the flask? (b) How many grams of ethane remain in the flask?

18.59 •• CP A balloon of volume 750 m^3 is to be filled with hydrogen at atmospheric pressure ($1.01 \times 10^5 \text{ Pa}$). (a) If the hydrogen is stored in cylinders with volumes of 1.90 m^3 at a gauge pressure of $1.20 \times 10^6 \text{ Pa}$, how many cylinders are required? Assume that the temperature of the hydrogen remains constant. (b) What is the total weight (in addition to the weight of the gas) that can be supported by the balloon if both the gas in the balloon and the surrounding air are at 15.0°C ? The molar mass of hydrogen (H_2) is 2.02 g/mol . The density of air at 15.0°C and atmospheric pressure is 1.23 kg/m^3 . See Chapter 12 for a discussion of buoyancy. (c) What weight could be supported if the balloon were filled with helium (molar mass 4.00 g/mol) instead of hydrogen, again at 15.0°C ?

18.60 •• A vertical cylindrical tank contains 1.80 mol of an ideal gas under a pressure of 0.300 atm at 20.0°C . The round part of the tank has a radius of 10.0 cm , and the gas is supporting a piston that can move up and down in the cylinder without friction. There is a vacuum above the piston. (a) What is the mass of this piston? (b) How tall is the column of gas that is supporting the piston?

18.61 •• CP A large tank of water has a hose connected to it (Fig. P18.61). The tank is sealed at the top and has compressed air between the water surface and the top. When the water height h has the value 3.50 m , the absolute pressure p of the compressed air is $4.20 \times 10^5 \text{ Pa}$. Assume that the air above the water expands at constant temperature, and take the atmospheric pressure to be $1.00 \times 10^5 \text{ Pa}$. (a) What is the speed with which water flows out of the hose when $h = 3.50 \text{ m}$? (b) As water flows out of the tank, h decreases. Calculate the speed of flow for $h = 3.00 \text{ m}$ and for $h = 2.00 \text{ m}$. (c) At what value of h does the flow stop?

Figure P18.61



18.62 •• CP A light, plastic sphere with mass $m = 9.00 \text{ g}$ and density $\rho = 4.00 \text{ kg/m}^3$ is suspended in air by thread of negligible mass. (a) What is the tension T in the thread if the air is at 5.00°C and $p = 1.00 \text{ atm}$? The molar mass of air is 28.8 g/mol . (b) How much does the tension in the thread change if the temperature of the gas is increased to 35.0°C ? Ignore the change in volume of the plastic sphere when the temperature is changed.

18.63 •• BIO How Many Atoms Are You? Estimate the number of atoms in the body of a 50 kg physics student. Note that the human body is mostly water, which has molar mass 18.0 g/mol , and that each water molecule contains three atoms.

18.64 •• BIO A person at rest inhales 0.50 L of air with each breath at a pressure of 1.00 atm and a temperature of 20.0°C . The inhaled air is 21.0% oxygen. (a) How many oxygen molecules does this person inhale with each breath? (b) Suppose this person is now resting at an elevation of 2000 m but the temperature is still 20.0°C . Assuming that the oxygen percentage and volume per inhalation are the same as stated above, how many oxygen molecules does this person now inhale with each breath? (c) Given that the body still requires the same number of oxygen molecules per second as at sea level to maintain its functions, explain why some people report “shortness of breath” at high elevations.

18.65 •• You blow up a spherical balloon to a diameter of 50.0 cm until the absolute pressure inside is 1.25 atm and the temperature is 22.0°C . Assume that all the gas is N_2 , of molar mass 28.0 g/mol . (a) Find the mass of a single N_2 molecule. (b) How much translational kinetic energy does an average N_2 molecule have? (c) How many N_2 molecules are in this balloon? (d) What is the *total* translational kinetic energy of all the molecules in the balloon?

18.66 • The size of an oxygen molecule is about $2.0 \times 10^{-10} \text{ m}$. Make a rough estimate of the pressure at which the finite volume of the molecules should cause noticeable deviations from ideal-gas behavior at ordinary temperatures ($T = 300 \text{ K}$).

18.67 •• CP CALC The Lennard-Jones Potential. A commonly used potential-energy function for the interaction of two molecules (see Fig. 18.8) is the Lennard-Jones 6-12 potential:

$$U(r) = U_0 \left[\left(\frac{R_0}{r} \right)^{12} - 2 \left(\frac{R_0}{r} \right)^6 \right]$$

where r is the distance between the centers of the molecules and U_0 and R_0 are positive constants. The corresponding force $F(r)$ is given in Eq. (14.26). (a) Graph $U(r)$ and $F(r)$ versus r . (b) Let r_1 be the value of r at which $U(r) = 0$, and let r_2 be the value of r at which $F(r) = 0$. Show the locations of r_1 and r_2 on your graphs of $U(r)$ and $F(r)$. Which of these values represents the equilibrium separation between the molecules? (c) Find the values of r_1 and r_2 in terms of R_0 , and find the ratio r_1/r_2 . (d) If the molecules are located a distance r_2 apart [as calculated in part (c)], how much work must be done to pull them apart so that $r \rightarrow \infty$?

18.68 • (a) What is the total random translational kinetic energy of 5.00 L of hydrogen gas (molar mass 2.016 g/mol) with pressure $1.01 \times 10^5 \text{ Pa}$ and temperature 300 K ? (Hint: Use the procedure of Problem 18.65 as a guide.) (b) If the tank containing the gas is placed on a swift jet moving at 300.0 m/s , by what percentage is the *total* kinetic energy of the gas increased? (c) Since the kinetic energy of the gas molecules is greater when it is on the jet, does this mean that its temperature has gone up? Explain.

18.69 •• It is possible to make crystalline solids that are only one layer of atoms thick. Such “two-dimensional” crystals can be created by depositing atoms on a very flat surface. (a) If the atoms in such a two-dimensional crystal can move only within the plane of the crystal, what will be its molar heat capacity near room temperature? Give your answer as a multiple of R and in $\text{J/mol} \cdot \text{K}$. (b) At very low temperatures, will the molar heat capacity of a two-dimensional crystal be greater than, less than, or equal to the result you found in part (a)? Explain why.

18.70 • Hydrogen on the Sun. The surface of the sun has a temperature of about 5800 K and consists largely of hydrogen atoms. (a) Find the rms speed of a hydrogen atom at this temperature. (The mass of a single hydrogen atom is $1.67 \times 10^{-27} \text{ kg}$.) (b) The escape speed for a particle to leave the gravitational influence of the sun is given by $(2GM/R)^{1/2}$, where M is the sun’s mass, R its radius, and G the gravitational constant (see Example 13.5 of Section 13.3). Use Appendix F to calculate this escape speed. (c) Can appreciable quantities of hydrogen escape from the sun? Can *any* hydrogen escape? Explain.

18.71 •• CP (a) Show that a projectile with mass m can “escape” from the surface of a planet if it is launched vertically upward with a kinetic energy greater than mgR_p , where g is the acceleration due to gravity at the planet’s surface and R_p is the planet’s radius. Ignore air resistance. (See Problem 18.70.) (b) If the planet in question is the earth, at what temperature does the average translational kinetic energy of a nitrogen molecule (molar mass 28.0 g/mol) equal that required to escape? What about a hydrogen molecule (molar mass 2.02 g/mol)? (c) Repeat part (b) for the moon, for which $g = 1.63 \text{ m/s}^2$ and $R_p = 1740 \text{ km}$. (d) While the earth and the moon have similar average surface temperatures, the moon has essentially no atmosphere. Use your results from parts (b) and (c) to explain why.

18.72 •• Helium gas is in a cylinder that has rigid walls. If the pressure of the gas is 2.00 atm , then the root-mean-square speed of the helium atoms is $v_{\text{rms}} = 176 \text{ m/s}$. By how much (in atmospheres) must the pressure be increased to increase the v_{rms} of the He atoms by 100 m/s ? Ignore any change in the volume of the cylinder.

18.73 •• CALC Calculate the integral in Eq. (18.31), $\int_0^\infty v^2 f(v) dv$, and compare this result to $(v^2)_{\text{av}}$ as given by Eq. (18.16). (*Hint:* You may use the tabulated integral

$$\int_0^\infty x^{2n} e^{-\alpha x^2} dx = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2^{n+1} \alpha^n} \sqrt{\frac{\pi}{\alpha}}$$

where n is a positive integer and α is a positive constant.)

18.74 •• (a) Calculate the total *rotational* kinetic energy of the molecules in 1.00 mol of a diatomic gas at 300 K. (b) Calculate the moment of inertia of an oxygen molecule (O_2) for rotation about either the y - or z -axis shown in Fig. 18.18b. Treat the molecule as two massive points (representing the oxygen atoms) separated by a distance of 1.21×10^{-10} m. The molar mass of oxygen *atoms* is 16.0 g/mol. (c) Find the rms angular velocity of rotation of an oxygen molecule about either the y - or z -axis shown in Fig. 18.18b. How does your answer compare to the angular velocity of a typical piece of rapidly rotating machinery (10,000 rev/min)?

18.75 •• CALC (a) Explain why in a gas of N molecules, the number of molecules having speeds in the *finite* interval v to $v + \Delta v$ is $\Delta N = N \int_v^{v+\Delta v} f(v) dv$. (b) If Δv is small, then $f(v)$ is approximately constant over the interval and $\Delta N \approx N f(v) \Delta v$. For oxygen gas (O_2 , molar mass 32.0 g/mol) at $T = 300$ K, use this approximation to calculate the number of molecules with speeds within $\Delta v = 20$ m/s of v_{mp} . Express your answer as a multiple of N . (c) Repeat part (b) for speeds within $\Delta v = 20$ m/s of $7v_{\text{mp}}$. (d) Repeat parts (b) and (c) for a temperature of 600 K. (e) Repeat parts (b) and (c) for a temperature of 150 K. (f) What do your results tell you about the shape of the distribution as a function of temperature? Do your conclusions agree with what is shown in Fig. 18.23?

18.76 •• CALC Calculate the integral in Eq. (18.30), $\int_0^\infty v f(v) dv$, and compare this result to v_{av} as given by Eq. (18.35). (*Hint:* Make the change of variable $v^2 = x$ and use the tabulated integral

$$\int_0^\infty x^n e^{-\alpha x} dx = \frac{n!}{\alpha^{n+1}}$$

where n is a positive integer and α is a positive constant.)

18.77 •• CP Oscillations of a Piston.

A vertical cylinder of radius r contains an ideal gas and is fitted with a piston of mass m that is free to move (Fig. P18.77). The piston and the walls of the cylinder are frictionless, and the entire cylinder is placed in a constant-temperature bath. The outside air pressure is p_0 . In equilibrium, the piston sits at a height h above the bottom of the cylinder. (a) Find the absolute pressure of the gas trapped below the piston when in equilibrium. (b) The piston is pulled up by a small distance and released. Find the net force acting on the piston when its base is a distance $h + y$ above the bottom of the cylinder, where $y \ll h$. (c) After the piston is displaced from equilibrium and released, it oscillates up and down. Find the frequency of these small oscillations. If the displacement is not small, are the oscillations simple harmonic? How can you tell?

18.78 •• DATA A steel cylinder with rigid walls is evacuated to a high degree of vacuum; you then put a small amount of helium into the cylinder. The cylinder has a pressure gauge that measures the pressure of the gas inside the cylinder. You place the cylinder in various temperature environments, wait for thermal equilibrium to be established, and then measure the pressure of the gas. You obtain these results:

	T ($^{\circ}\text{C}$)	p (Pa)
Normal boiling point of nitrogen	-195.8	254
Ice–water mixture	0.0	890
Outdoors on a warm day	33.3	999
Normal boiling point of water	100.0	1214
Hot oven	232	1635

(a) Recall (Chapter 17) that absolute zero is the temperature at which the pressure of an ideal gas becomes zero. Use the data in the table to calculate the value of absolute zero in $^{\circ}\text{C}$. Assume that the pressure of the gas is low enough for it to be treated as an ideal gas, and ignore the change in volume of the cylinder as its temperature is changed. (b) Use the coefficient of volume expansion for steel in Table 17.2 to calculate the percentage change in the volume of the cylinder between the lowest and highest temperatures in the table. Is it accurate to ignore the volume change of the cylinder as the temperature changes? Justify your answer.

18.79 ••• DATA The Dew Point

and Clouds. The vapor pressure of water (see Exercise 18.46) decreases as the temperature decreases. The table lists the vapor pressure of water at various temperatures:

If the amount of water vapor in the air is kept constant as the air is cooled, the *dew point* temperature is reached, at which the partial pressure and vapor pressure coincide and the vapor is saturated. If the air is cooled further, the vapor condenses to liquid until the partial pressure again equals the vapor pressure at that temperature. The temperature in a room is 30.0°C .

Temperature ($^{\circ}\text{C}$)	Vapor Pressure (Pa)
10.0	1.23×10^3
12.0	1.40×10^3
14.0	1.60×10^3
16.0	1.81×10^3
18.0	2.06×10^3
20.0	2.34×10^3
22.0	2.65×10^3
24.0	2.99×10^3
26.0	3.36×10^3
28.0	3.78×10^3
30.0	4.25×10^3

(a) A meteorologist cools a metal

can by gradually adding cold water. When the can's temperature reaches 16.0°C , water droplets form on its outside surface. What is the relative humidity of the 30.0°C air in the room? On a spring day in the midwestern United States, the air temperature at the surface is 28.0°C . Puffy cumulus clouds form at an altitude where the air temperature equals the dew point. If the air temperature decreases with altitude at a rate of $0.6^{\circ}\text{C}/100$ m, at approximately what height above the ground will clouds form if the relative humidity at the surface is (b) 35%; (c) 80%?

18.80 •• DATA The statistical quantities “average value” and “root-mean-square value” can be applied to any distribution. **Figure P18.80** shows the scores of a class of 150 students on a 100 point quiz. (a) Find the average score for the class. (b) Find the rms score for the class. (c) Which is higher: the average score or the rms score? Why?

Figure P18.80

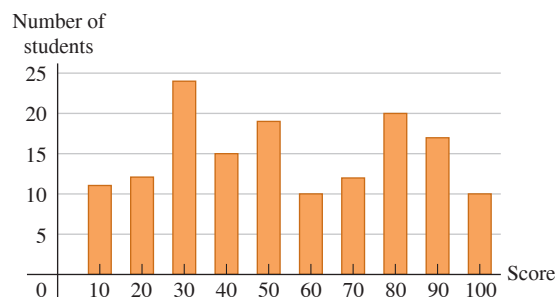
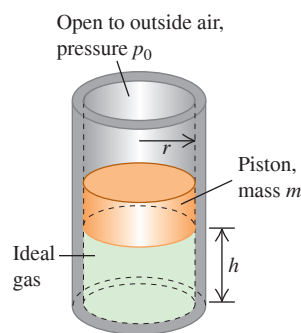
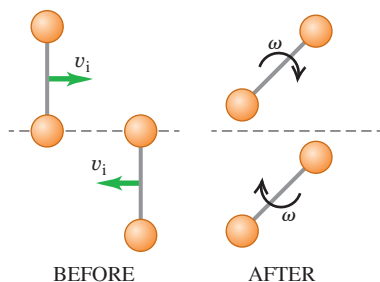


Figure P18.77



18.81 ••• CP We can crudely model a nitrogen molecule as a pair of small balls, each with the mass of a nitrogen atom, 2.3×10^{-26} kg, attached by a rigid massless rod with length $d = 2r = 188$ pm. (a) What is the moment of inertia of this molecule with respect to an axis passing through the midpoint and perpendicular to the molecular axis? (b) Consider air at 1 atm pressure and 20.0°C temperature. Suppose two nitrogen molecules with rms speeds v_i have an elastic collision such that only one nitrogen atom in one molecule collides with one nitrogen atom in the other molecule in the manner depicted in **Fig. P18.81**. Write equations for the conservation of energy and the conservation of angular momentum, in terms of r , v_i , v_f , and ω , where v_f and ω are, respectively, the center-of-mass speed and the angular speed of either molecule after the collision. (c) Solve these equations for v_f and ω in terms of v_i and r . Note that $\omega \neq 0$ after the collision. (d) Using $v_i = v_{\text{rms}}$ for the conditions specified above, what is the frequency of rotation?

Figure P18.81



CHALLENGE PROBLEMS

18.82 ••• A pneumatic lift consists of a vertical cylinder with a radius of 10.0 cm. A movable piston slides within the cylinder at its upper end and supports a platform on which loads are placed. An intake valve allows compressed air from a tank to enter the cylinder, and an exhaust valve allows air to be removed from the cylinder. In either case the rate of air transfer is sufficiently low that the temperature in the cylinder remains constant. When neither valve is activated, the cylinder is airtight. The piston and platform together have a mass of 50.0 kg, the temperature is 20.0°C , and the pressure outside the cylinder is 1.00 atm. (a) There is 1.00 mol of air in the cylinder and no load on the platform. What is the height h between the bottom of the piston and the bottom of the cylinder? (b) A 200 kg load is placed on the platform. By what distance does the platform drop? (c) The intake valve is activated and compressed air enters the cylinder so that the platform moves back to its original height. How many moles of air were introduced? (d) How many more moles of air should be introduced so that the platform rises 2.00 m above its original height? (e) At what rate should this air be introduced, in mmol/s, so that the platform rises at a rate of 10.0 cm/s?

18.83 ••• CP Dark Nebulae and the Interstellar Medium. The dark area in **Fig. P18.83** that appears devoid of stars is a *dark nebula*, a cold gas cloud in interstellar space that contains enough material to block out light from the stars behind it. A typical dark nebula is about 20 light-years in diameter and contains about 50 hydrogen atoms per cubic centimeter (monatomic hydrogen, *not* H_2) at about 20 K. (A light-year is the distance light travels in vacuum in one year and is equal to 9.46×10^{15} m.) (a) Estimate the mean free path for a hydrogen atom in a dark nebula. The radius of a hydrogen atom is 5.0×10^{-11} m. (b) Estimate the rms speed of a hydrogen atom and the mean free time (the average time between collisions for a given atom). Based on this result, do you think that atomic collisions, such as those leading to H_2 molecule formation, are very important in determining the composition of

the nebula? (c) Estimate the pressure inside a dark nebula. (d) Compare the rms speed of a hydrogen atom to the escape speed at the surface of the nebula (assumed spherical). If the space around the nebula were a vacuum, would such a cloud be stable or would it tend to evaporate? (e) The stability of dark nebulae is explained by the presence of the *interstellar medium* (ISM), an even thinner gas that permeates space and in which the dark nebulae are embedded. Show that for dark nebulae to be in equilibrium with the ISM, the numbers of atoms per volume (N/V) and the temperatures (T) of dark nebulae and the ISM must be related by

$$\frac{(N/V)_{\text{nebula}}}{(N/V)_{\text{ISM}}} = \frac{T_{\text{ISM}}}{T_{\text{nebula}}}$$

(f) In the vicinity of the sun, the ISM contains about 1 hydrogen atom per 200 cm^3 . Estimate the temperature of the ISM in the vicinity of the sun. Compare to the temperature of the sun's surface, about 5800 K. Would a spacecraft coasting through interstellar space burn up? Why or why not?

Figure P18.83



18.84 ••• CALC Earth's Atmosphere. In the *troposphere*, the part of the atmosphere that extends from earth's surface to an altitude of about 11 km, the temperature is not uniform but decreases with increasing elevation. (a) Show that if the temperature variation is approximated by the linear relationship

$$T = T_0 - \alpha y$$

where T_0 is the temperature at the earth's surface and T is the temperature at height y , the pressure p at height y is

$$\ln\left(\frac{p}{p_0}\right) = \frac{Mg}{R\alpha} \ln\left(\frac{T_0 - \alpha y}{T_0}\right)$$

where p_0 is the pressure at the earth's surface and M is the molar mass for air. The coefficient α is called the lapse rate of temperature. It varies with atmospheric conditions, but an average value is about $0.6^\circ\text{C}/100 \text{ m}$. (b) Show that the above result reduces to the result of Example 18.4 (Section 18.1) in the limit that $\alpha \rightarrow 0$. (c) With $\alpha = 0.6^\circ\text{C}/100 \text{ m}$, calculate p for $y = 8863 \text{ m}$ and compare your answer to the result of Example 18.4. Take $T_0 = 288 \text{ K}$ and $p_0 = 1.00 \text{ atm}$.

MCAT-STYLE PASSAGE PROBLEMS

Insulating Windows. One way to improve insulation in windows is to fill a sealed space between two glass panes with a gas that has a lower thermal conductivity than that of air. The thermal conductivity k of a gas depends on its molar heat capacity C_V , molar mass M , and molecular radius r . The dependence on those quantities at a given temperature is approximated by $k \propto C_V/r^2\sqrt{M}$. The noble gases have properties that make them particularly good choices as insulating gases. Noble gases range from helium (molar mass 4.0 g/mol, molecular radius 0.13 nm) to xenon (molar mass 131 g/mol, molecular radius 0.22 nm). (The noble gas radon is heavier than xenon, but radon is radioactive and so is not suitable for this purpose.)

18.85 What is one reason the noble gases are *preferable* to air (which is mostly nitrogen and oxygen) as an insulating material? (a) Noble gases are monatomic, so no rotational modes contribute to their molar heat capacity; (b) noble gases are monatomic, so they

have lower molecular masses than do nitrogen and oxygen; (c) molecular radii in noble gases are much larger than those of gases that consist of diatomic molecules; (d) because noble gases are monatomic, they have many more degrees of freedom than do diatomic molecules, and their molar heat capacity is reduced by the number of degrees of freedom.

18.86 Estimate the ratio of the thermal conductivity of Xe to that of He. (a) 0.015; (b) 0.061; (c) 0.10; (d) 0.17.

18.87 The rate of *effusion*—that is, leakage of a gas through tiny cracks—is proportional to v_{rms} . If tiny cracks exist in the material that's used to seal the space between two glass panes, how many times greater is the rate of He leakage out of the space between the panes than the rate of Xe leakage at the same temperature? (a) 370 times; (b) 19 times; (c) 6 times; (d) no greater—the He leakage rate is the same as for Xe.

ANSWERS

Chapter Opening Question ?

(iv) From Eq. (18.19), the root-mean-square speed of a gas molecule is proportional to the square root of the absolute temperature T . The temperature range we're considering is $(25 + 273.15) \text{ K} = 298 \text{ K}$ to $(100 + 273.15) \text{ K} = 373 \text{ K}$. Hence the speeds increase by a factor of $\sqrt{(373 \text{ K})/(298 \text{ K})} = 1.12$; that is, there is a 12% increase. While 100°C feels far warmer than 25°C , the difference in molecular speeds is relatively small.

Key Example VARIATION Problems

VP18.4.1 (a) $4.67 \times 10^5 \text{ Pa}$ (b) 0.278 mol

VP18.4.2 (a) $1.20 \times 10^3 \text{ m}^3$, (b) 4.52 times

VP18.4.3 (a) $2.9 \times 10^{-3} \text{ mol/m}^3$ (b) $8.0 \times 10^{-5} \text{ kg/m}^3$

VP18.4.4 (a) $T_2 = T_1 \left(\frac{p_2}{p_1} \right)^{2/5}$ (b) $\frac{p_2}{p_1} = 0.660$, $\frac{T_2}{T_1} = 0.758$

(c) $\frac{p_2}{p_1} = 1.52$, $\frac{T_2}{T_1} = 1.32$

VP18.7.1 -231°C

VP18.7.2 (a) $6.2 \times 10^{-20} \text{ J}$ (b) $8.6 \times 10^3 \text{ m/s}$

VP18.7.3 (a) 1.5×10^{27} (b) $9.1 \times 10^6 \text{ J}$

(c) $1.1 \times 10^2 \text{ m/s}$, or $4.0 \times 10^2 \text{ km/h}$

VP18.7.4 (a) 5.50 (b) 38.5 (c) 6.20

VP18.8.1 (a) $6.8 \times 10^{-6} \text{ m}$ (b) 1.2×10^2 times greater

VP18.8.2 (a) $5.7 \times 10^{-3} \text{ Pa}$ (b) $2.3 \times 10^{-6} \text{ mol}$

VP18.8.3 (a) $1.2 \times 10^{-8} \text{ m}$ (b) 8.9×10^{-12}

VP18.8.4 (a) $t = \frac{1}{4\pi r^2 p} \sqrt{\frac{mkT}{6}}$ (b) doubling r

Bridging Problem

(a) 102 km

(b) yes